Competition and the Cost of U.S. Infrastructure

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Abstract

Can limited competition in procurement auctions explain the high and rising price of road infrastructure in the U.S.? I assemble a new dataset covering the near-universe of state highway auctions between 2002 and 2024. I first document thin competition: one- or two-bidder auctions account for a third of awards, and this share has risen over the past decade. Using spatial variation in inter-state bidder locations, I then estimate the average causal effect of competition on prices; an additional bidder reduces prices by ten percent. To decompose bids into markups and production costs, I develop a semi-parametric structural auction model that incorporates bidders' uncertainty over the number of competitors they face. I show that price increases over the past decade are more attributable to increasing markups than increasing production costs. Limited competition, in turn, is consistent with patterns generated by fixed costs of entering public procurement, but not broad construction-sector fixed costs. Embedding the markup estimates in a structural entry model, I estimate large auction-level—and even larger market-level—entry costs, suggesting an important role for procurement bureaucracy and regulatory barriers.

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1 Introduction

American infrastructure is remarkably expensive. For example, the per-mile price paid by the U.S. government to construct transit lines can be up to double that paid by peer countries.[1] Despite extensive public attention, the underlying causes of these high prices remain a puzzle.[2] One longstanding view is that high prices reflect high production costs, driven for example by high labor costs or low productivity (Kessler and Katz, 1999; Goolsbee and Syverson, 2023). However, since governments typically procure infrastructure from private firms that compete for public contracts, often via auctions, high prices may also be driven by limited competition and resulting firm markups over production costs. If those markups are large, however, their persistence raises a further question: why do firms not enter the market and compete them away?

The view that competition is an important driver of infrastructure prices is notably present in free-response answers from transportation procurement officials surveyed by Liscow et al. (2023), where the most commonly cited cost driver is "Competition," followed by "Materials and Labor." Disentangling the contributions of markups versus production costs is vital to increasing the direct return on infrastructure investment, and thus realizing the well-documented benefits of better infrastructure on trade, productivity, labor markets, commuting and mobility, and consumption.[3]

At least three factors make it challenging to disentangle these channels in practice. First, fragmented procurement systems across the country have resulted in scarce data, such that even simple statistics — such as the number of bidders per procurement auction — are difficult to obtain. Second, infrastructure projects differ substantially in scope and complexity, making projects difficult to compare across years and across markets. Third, simple statistics do not map cleanly to conclusions about the effects of competition: estimating these effects requires understanding firms' strategic behavior and their information sets — accounting, for example, for firms' limited information about

[1] The average per-km cost of U.S. projects is 33%, 103%, 200%, and 680% higher than for the U.K., Australia, Canada, and Spain, respectively, according to data compiled by the NYU Marron Institute. See Figure A.1 for a plot of the data.

[2] See Smith (2017) and Vartabedian (2021) for examples of news articles hypothesizing on cost drivers.

[3] For trade see Limão and Venables (2001), for productivity see Donaldson and Hornbeck (2016), Ghani et al. (2014), Baum-Snow (2014), Gibbons, Lyytikäinen, et al. (2019), for labor markets see Chandra and Thompson (2000), Michaels (2008), for commuting and mobility see Gibbons and Machin (2005), Glaeser et al. (2008), and for consumption see Gonzalez-Navarro and Quintana-Domeque (2016), among others.

the number of competitors they face in any given auction. Ignoring these uncertainties biases estimates of markups.

In this paper, I overcome these challenges in the context of highway infrastructure auctions in the continental U.S. Nearly all major roadwork and bridges fall into this category, and this procurement method embodies the most common form of infrastructure procurement worldwide.[4] I first assemble a novel dataset of 1.3 million project-level bids spanning two decades of auctions, and I use this data to document novel facts about costs and competition in road procurement: most importantly, that prices are rising and competition is strikingly thin and decreasing over time. I exploit spatial variation in inter-state bidder locations to estimate the average causal effect of competition on winning bids and find that an additional bidder reduces prices by ten percent. I then develop and estimate a semi-parametric structural model of auction bidding that is able to decompose bids into markups and production costs. The model accounts for firms' uncertainty over the entry decisions of other bidders and the government's reserve price, and is estimated with minimal functional form assumptions, leveraging the richness of the data. The model estimates reveal the role of markups in historical price trends, while a model counterfactual quantifies price reductions from hypothetical increases in participation. Large markups, however, raise the question of why firms do not enter and compete them away. I address this by nesting the bidding model within an entry model. The bidding model recovers firms' expected variable profits, while observed entry decisions reveal the fixed costs they must overcome. This allows me to speak not only to the *consequences*, but also the *causes*, of limited competition. I find an important role for bureaucratic costs of entering auctions and even larger costs of entering new state markets, pointing to state-specific procurement and regulatory barriers as a central source of limited competition.

I begin by assembling panel data on the near universe of auctions for state Departments of Transportation (DOTs) from 2002 to 2024. States are the principal governmental unit responsible for road construction, and the vast majority of projects are awarded through sealed-bid first price auctions. For each auction, I have a series of covariates, including a measure of size and a brief project description. Importantly, for a sub-sample of ten years, I have the bid-schedule for each project, which is the complete list of construction inputs. This data substantially extends prior

[4] Of 40 countries surveyed by the World Bank, all primarily used design-bid-build methods with competitive bidding for traditional infrastructure investment (World Bank, 2020).

sources, which in the empirical auction literature typically cover only a single state or a small handful of states.[5]

The data reveals new facts on the U.S. road procurement market, in particular rising prices and weakening competition. I find that over the past two decades, inflation-adjusted winning bids have more than doubled. The trend persists through a variety of sub-samples of the data, such as restricting to similarly sized projects, and the price of important inputs do little to explain it. By contrast, within-state price trends from two-way-fixed-effects regressions are correlated with changes in variables related to market structure, such as the number of bidders. To my knowledge, this is the first academic study of 21st-century price growth. These results extend Brooks and Liscow (2023), who show the Interstate Highway System spending per new mile tripled in real terms in the 1960s–1980s, finding continued growth despite the shift from new construction to reconstruction.[6]

To probe the mechanisms underlying high prices, I start by providing simple measures of competition. I document that, nationwide and over time, transportation auctions have attracted only a small pool of bidders. The median auction draws just three participants, and one- or two-bidder auctions account for about a third of all projects. Since 2010, the share of auctions with 1–3 bidders increased from 47% to 63% by 2024. Declining participation occurs against a backdrop of minimal market entry and stable demand, despite the billions of dollars allocated towards transportation infrastructure in the 2021 Infrastructure Investment and Jobs Act (IIJA). The phenomenon of few bidders in this setting echoes a broader pattern in government procurement. For example, Kang and Miller (2021) document that a large share of federal IT procurement contracts attract only a single bidder, reflecting demand-side choices as well as supply factors.[7] The data in this study provide a newly comprehensive picture of low bidder participation in transportation procurement, where contracts are intended to draw high competition.

[5] Liscow et al. (2023) provide the first data spanning all states through the combination of survey and public record requests; however, the authors' bid analysis is limited to five winning bids per state. My dataset additionally captures the full distribution of bids across the continental U.S.

[6] Brooks and Liscow (2023) show suggestive evidence that the 20th century price growth is caused by permitting costs and citizen voice frictions, which are likely to be more binding for brand new construction than for the reconstruction projects that make up modern roadwork.

[7] In the authors' setting, a low number of bidders reflects several factors. First, the federal government procures many projects without competition. Second, even when contracts are classified as competitive, procurement is primarily conducted through negotiation rather than auction. The authors argue that, conditional on potential bidders being similar, the government can successfully extract maximal rents through negotiation without the need for additional bidders.

It is difficult to assess the impact of an additional bidder on price from descriptive data alone. However, the gap between the first and second lowest bids in an auction, provides a simple measure of the loss to government if the first bidder were to drop out. I find this bid gap declines convexly as the number of bidders increases, revealing that returns to competition hinge, intuitively, on the baseline number of bidders. However, this measure fails to capture two effects that determine the magnitude of those returns: a *composition* effect, where the marginal bidder is the low bidder some probability , and a *strategy* effect, where other firms adjust their bids in response to more competition. The composition and strategy effects together determine the total causal effect.

Consequently, the second contribution of this paper is to directly estimate the average causal effect of an additional bidder on auction prices. Theoretical auction models often predict large returns, but direct empirical evidence is limited and more mixed.[8,9] The primary identification challenge is selection into auctions, which induces a positive correlation between the number of bidders and the error term in the price equation. I address this with an instrumental variables strategy that exploits two sources of variation. I identify inter-state entry by firms based outside the market of interest; such entry is rare but lumpy, and when it occurs firms typically enter many auctions, creating plausibly exogenous and salient variation in local competition. I interact state entry with the distance to the out-of-state bidder's establishment, as distance is a strong predictor of a firm auction entry. My estimates suggest that on average an additional bidder lowers the winning bid by 10%.

I turn to a structural auction model of bidding in order to decompose the prices paid by the government into two pieces: markups and firms' variable production costs. The model allows me to (i) estimate the average markup and heterogeneity, for example, across different baseline number of bidders; (ii) estimate whether declining competition can explain the historical rise in prices; and (iii) quantify counterfactual returns from increased bidder competition in the future.

I build on an independent private values auction model, incorporating key features of the setting. In particular, while competition is central, firms do not know the exact number of competitors they face in an auction. Deviating from the standard literature, I instead model firms as knowing how likely they are to face competition. Specifically, firms observe the probability distribution of the number

[8]For example, Bulow and Klemperer (1996) show that under standard assumptions, increasing the number of bidders by one benefits the seller more than adopting the optimal mechanism would.

[9]For two papers on this topic, see Coviello and Mariniello (2014) and Lewis-Faupel et al. (2016).

of bidders in an auction.[10] I model the government's discretion to reject bids as a secret reserve price, where again, bidders have information on the distribution but not the realization. One benefit to the researcher of the auction setting is that, under independence assumptions and information assumptions like the above, a profit maximizing firm has an optimal bid given their cost draw. I take advantage of this to identify the unobserved production costs from observed bids with minimal functional form assumptions, following Guerre et al. (2000).

I estimate the model for all roadwork auctions in the continental U.S. between 2014 and 2023, the period for which I have the complete bid-schedules for each project. I split the estimation by state, year, and a covariate-based cluster, allowing trends over time and across states to be fully flexible. Joint auction estimation within samples requires accounting for project heterogeneity, which I address by "homogenizing" bids to remove the effects of project covariates, as in Haile et al. (2003). Building on the traditional linear regression method, I use machine learning on covariates and absorb residual unobserved heterogeneity with an auction random effect.

The structural estimates indicate substantial markups. The mean markup is around 0.2. This corresponds to an average per-auction margin of a bit more than $0.7 million. A back-of-the-envelope calculation yields a total of roughly $25 billion spent above production cost over the ten year period. While most states exhibit sizable markups, there is large dispersion across states: moving from the 25th to the 75th percentile raises the markup by 37%.

I examine whether the rise in price between 2014 and 2023 can be explained by decreasing competition. I find that roughly one third of the 28% realized price increase over the ten year period is accounted for by the project observables, suggesting that even within project type, projects are becoming more complex. More complex projects require more inputs in total and may require novel inputs, such as environmental adjustments. Of the remaining two-thirds growth that is real price growth, costs explain little, rising only 8%. Because this measure captures all variable costs borne by the firm—including materials, labor, and opportunity costs—the estimates imply relatively little growth in aggregate production costs. Increasing margins explain the majority of the real price growth, rising by 45% over the period. Changing competition, rather than production costs, appears

[10] A few papers incorporate uncertainty over the number of bidders, but differ in framework. For instance, Li and Zheng (2009) and De Silva and Rosa (2024) model uncertainty by explicitly modeling potential entrants in mixed-strategy entry equilibria, in particular relying on parametric cost distributions.

to be the defining feature of infrastructure over the last decade.

How much could the government save with future higher bidder participation? I solve for counterfactual prices in the model using the most recent distribution of costs but under the hypothetical guarantee of one additional bidder per auction. The counterfactual is able to account for the current distribution of number of bidders — an advantage over the reduced form analysis — and decomposes the total effect into the composition effect and the strategy effect. I find an average savings of $350,000 per project, remarkably close to the 10% estimated in the reduced form exercise, although the identifying assumptions have no overlap. The total effect is driven primarily by the strategic effect, suggesting more productive entrants is a limited channel for price reduction.

Markups are large and persistent, yet firms do not enter auctions and drive down these variable profits. In the final section, I turn from measuring the consequences of low bidder participation to studying the causes. I present evidence that fixed costs are substantial, deterring entry. I rule out construction capital requirements, instead showing evidence of significant entry barriers at both the auction and the state market level. These barriers are consistent with procurement requirements playing a central role in limiting participation, in line with views that decades of accrued rules have led to policy congestion (Klein and Thompson, 2024). Entry costs also imply returns to market scale: both larger auctions and thicker demand would reduce markups.

If policymakers aim to address entry costs, it is useful to know where the bottlenecks lie. The tasks of both measuring regulation and identifying root causes in layered bureaucracy is complex (Trebbi and Zhang, 2022, Bartik et al., 2025). Instead, I estimate the magnitudes of market entry and auction entry costs and compare the two. To do so, I combine the results from my bidding model with an auction entry model and, for estimation, exploit the fact that firms enter only when expected total profits are nonnegative. The model builds off of Berry (1992), incorporating firm and project specific shocks and relying on an ordered entry assumption to ensure a unique equilibrium. The estimation nests the results from the bidding model. While more stylized than the bidding model, it yields a straightforward parameterization of auction entry costs.

I find sizable costs of entry into auctions, indicating that bureaucratic barriers to bidding are meaningful. Entry costs into state markets are much larger, suggesting that broader regulatory barriers play a more important role. While the infrastructure market sits alongside an enormous private

construction sector, and many firms could in principle participate in government procurement, for most the cost of entry appears prohibitive. Regulations generate important benefits, the measurement of which warrants future work. At the same time, this work suggests that policies that lower regulatory entry costs, or increase market scale, would meaningfully increase competition and reduce prices.

Related Literature

This paper's main contribution is to shed light on the competitive structure of the United States highway infrastructure market and its consequences. Other work seeking to understand infrastructure costs includes Brooks and Liscow (2023) and Mehrotra et al. (2024), finding evidence that citizen voice and material input costs contributed to higher costs in the late twentieth century and in 1984–2008, respectively. Most related to this paper is Liscow et al. (2023), who combine survey data with public records requests on highway projects and finds that low state capacity plays an important role in increasing project costs. The authors also report suggestive evidence that fewer bidders results in higher prices, a correlation further investigated here. While the aim of this project is to quantify the competition channel, state capacity problems like those in Liscow et al. (2023) may operate in the background, affecting either firms' production costs or their decision to work with the government. Other work focuses on other areas of government procurement, notably Kang and Miller (2021), who document extremely limited competition in federal procurement and argue that, in some settings, negotiated contracts with few bidders may be optimal. Looking outside the U.S., Kirchberger and Beirne (2021) study competition in the global construction industry with micro-data, showing that limited competition in cement generates substantial markups.

Several papers investigate distinct but intertwined questions about infrastructure. Goolsbee and Syverson (2023) study the deterioration of productivity in the U.S. construction sector, an issue that compounds rising costs. Viewed through their framework of baseline productivity growth, the question becomes why infrastructure prices have not fallen over time. Currier et al. (2023) study the quality of local road infrastructure and estimate the distribution of the costs of poor road quality across places. Kroft et al. (2025) study the relationship between construction production markups and monopsony power over construction workers.[11] Finally, Fajgelbaum et al. (2023) study the

[11]The authors capture the public market contribution with auction data from 28 states.

political incentives surrounding California's high-speed rail, a famously expensive megaproject, comparing welfare quantifications from a spatial model to observed voting patterns.

To model procurement auctions, I build on a rich empirical literature. Since Paarsch (1992), economists have inferred bidders' valuations from observed bids using the structure implied by optimal bidding strategies. Of particular relevance is the work on transportation infrastructure auctions, which includes the following. Bajari (1997) analyzes bidder asymmetry in Minnesota roadwork auctions. Hong and Shum (2002) examine winner's curse in New Jersey procurement. Jofre-Bonet and Pesendorfer (2003) estimate a dynamic capacity-constrained model in California. De Silva, Dunne, et al. (2003) and De Silva, Kosmopoulou, et al. (2009) document incumbent–entrant asymmetry in Oklahoma, while Li and Zheng (2009) study entry in Texas mowing contracts. Krasnokutskaya (2011) account for unobserved heterogeneity in Michigan highway auctions, and Krasnokutskaya and Seim (2011) evaluate small-firm preference policies in California. Lewis and Bajari (2011) analyzes scoring auctions in California; Bhattacharya et al. (2014) quantify entry-rights effects in bridge building in Oklahoma and Texas; and Jeziorski and Krasnokutskaya (2016) study subcontracting. Balat (2017) examine ARRA impacts on capacity and prices in California. Bolotnyy and Vasserman (2023) evaluate scaling auctions for bridge maintenance in Massachusetts. De Silva and Rosa (2024) link the Great Recession's private-sector downturn to lower Texas road prices, and Ito (2024) show that firms in Montana signal entry intent through online questions. These papers, and others, elucidate information, behavior, and optimal mechanism design in procurement auctions. I build on this work with a parsimonious model that preserves key features of the data and limits functional-form assumptions. I introduce a tractable way to account for bidder uncertainty over the number of competitors while recovering costs and markups semiparametrically before estimating entry. My treatment of bidder uncertainty relates to Li and Zheng (2009) and Jovanovic and Menkveld (2022), which also allow bidders to be uncertain about realized competition. Unlike those papers, I estimate bidding and entry separately rather than jointly modeling the two stages. A second difference is that additional realized competition weakly reduces prices in my framework, consistent with the reduced-form evidence in my setting. The tractability of my setup allows the model to be estimated at national scale. To my knowledge, no paper yet has collected national project-level infrastructure data and used the auction structure to evaluate markups in the U.S. as a whole. The combination of national data and this framework lets me trace limited competition from

observed bidder counts to markups, counterfactual gains from greater participation, and the entry costs that sustain thin markets.

2 Setting and Data

2.1 Transportation Infrastructure Market Structure

Major roads and bridges in the U.S. are built and maintained by state governments, using a mix of state and federal funding. Road spending is substantial: across federal, state, and local governments, the U.S. spends roughly $120 billion in a regular year on the maintenance and development of roads and bridges.[12] Recent attention to infrastructure has brought additional federal funding, including the $1.2 trillion Bipartisan Infrastructure Investment and Jobs Act (IIJA) of 2021, with roads and bridges receiving the largest share of spending.[13] Nearly all state roadwork is procured via auctions, with exceptions being along the lines of atypical large projects that require technical innovation and therefore outsourced design.[14] The typical project can be thought of as a mid-sized, routine project involving limited design innovation, such as the reconstruction of a segment of highway.

The vast majority of projects are completed through the following process.[15] First, project specifications are designed in-house by the state Department of Transportation (DOT). Then the contract for completing the project is awarded to a construction firm via a sealed-bid first price scaling auction. In a scaling auction, bidders are given detailed project specifications and a list of input item quantities, known as the bid-schedule. A bid consists of a price per bid-schedule item. The total price of the project is calculated, and the contract is awarded to the bidder with the lowest total.

Bolotnyy and Vasserman (2023) show that firms often bid strategically on item prices, underpricing

[12] Data from the Organization for Economic Co-operation and Development (OECD) Infrastructure Investment Survey (2024). The OECD spending figure is not broken down by level of government. Local governments are responsible for work done to local roads, which account for 77% of the road system mileage but only 13% of vehicle miles traveled (FHA, 2000).

[13] Roads and bridges were allocated $110 billion.

[14] An example of this extreme is Boston's Big Dig project – the most expensive highway project in U.S. history – awarded through negotiation to Bechtel Corporation, a company with an annual revenue of $16 billion and operate at a different scale than most roadwork firms.

[15] Alternative procurement methods account for fewer than 5% of DOT projects Liu et al. (2022).

items for which they expect the government to have overestimated quantities and overpricing those they believe were underestimated. As a result, item-level bids are an unreliable measure of input costs. In this project, I focus on the total bid, which determines firms' expected profit, and set aside the nested choice of pricing the individual items in the bid-schedule. Consequently, the auctions can be treated as simple first price auctions. This allows me to focus on the role of competition in shaping total prices.

On the supply side, construction firms operate in the public sector, private sector, or both. Appendix Table A.9 profiles a random sample of five companies that compete for government contracts; their listed sub-industries span road construction, carbon capture, general building, and ship repair. All five also serve private clients. This makes sense as the private sector is larger overall: a back-of-the-envelope calculation using Kroft et al. (2020) suggests that public-sector participants account for about 22% of sales in the construction industry.[16] Appendix Figure A.2 compares the distribution of sub-industries (four-digit SIC codes) for auction participants with those of all firms within two-digit SIC categories Building Construction and Heavy Construction.[17] Single-Family Housing Construction alone accounts for 78% of all firms, reflecting both the dominance of this sector and the smaller scale of typical firms. Housing remains the largest category among auction participants, but the distribution is markedly flatter, with a similar number of firms in Highway and Street Construction. Firms classified under housing construction in the auction sample usually operate across several sub-industries, as illustrated by the examples in Table A.9.

Public sector construction is not equivalent to private sector jobs. For example, firms must comply with government rules and procedures, covering everything from wage rates and inspection protocols to the formatting of correspondence. These requirements are largely determined by states and vary across jurisdictions. Pre-qualification prior to bidding can be demanding: among required certifications and documents, submissions can include fully written safety programs, trainings documentation, and reference letters on past performance.[18] Nearly all states also require proof of bonding capacity, a government-mandated form of insurance that is itself difficult to obtain.[19]

[16] Kroft et al. (2020) combine tax data on all U.S. firms with DOT auctions from 28 states. They estimate that firms participating in auctions represent 12% of sales among all firms coded as construction under NAICS.

[17] Data details are provided in Section 2.2 and Appendix B.2.

[18] For examples, see Maine and Georgia DOT prequalification documents.

[19] See Congressional Research Service Report "SBA Surety Bond Guarantee Program" Updated July 18, 2025 for a discussion of the difficulty in obtaining surety bonds, particularly to small businesses.

Once qualified, preparing an individual bid proposal remains itself resource-intensive: Liscow et al. (2023) report that the average submission is 164 pages of forms. If these pre-auction procedures are costly to firms, they may generate high equilibrium markups.

2.2 Data

Auction data My main data is state Department of Transportation procurement auction bids for all contiguous U.S. states. Observation start dates for states range from 1993 to 2007, but all are observed until 2024. I define the analysis window to begin in 2002, when 85% of states are observed.[20] The data come from a construction data software company; data compilation details, state idiosyncracies, and summary statistics of variables by state are reported in Appendix B.1.

In total, I observe 370,000 auctions with 1.3 million bids. An important feature of the setting is that I observe all bids, not just winning bids, which is a common limitation of auction datasets. The data covers the allocation of nearly 1.4 trillion (real) dollars spent by the U.S. government, accounting for roughly half of total U.S. road-related infrastructure spending.[21] The main excluded spending is roadwork conducted by local governments, primarily for local street repairs.

The data include winning bids but exclude any changes in price due to renegotiation. To gauge the magnitude of these changes, I collect separate data on change orders and final costs for four jurisdictions. Appendix Table A.1 reports mean and median cost overruns. The largest median percentage overrun is 3.25% in California, while the smallest is -3.87% in New York City, where projects actually come in under budget on average. Although renegotiation matters, baseline winning bids remain first-order.

The data also include firm names, which I use to match firms across states.[22] The details of this match process are in Appendix B.2. Additional variables included in the data are the project date, the county of the project, the engineering type (defined as the input category with the largest share of spending), and "tons," the sum of all amounts of bid-schedule items measured in tons. The latter

[20] When relevant, I restrict to a balanced panel or report balanced panel results in the appendix.

[21] The data documents $66 billion spent in 2023 via DOT auctions, while all levels of U.S. government combined spent approximately twice as much, $127 billion, on road-related infrastructure, according to the OECD survey "Investment Spending in Transport Infrastructure".

[22] I assume each firm name uniquely identifies a firm. This assumption appears reasonable based on the validation exercise. The main challenge arises from cross-state formatting differences.

is the best available measure of project size. I have a brief description of the project, which I use to assign each project to one of nine categories with a large language model (LLM).[23] Details on the classification procedure are in Appendix B.3.

Finally, I have the full set of project bid-schedule items from 2014-2023. This is the complete list of input items, units, and quantities for every project. The items are highly specific; an example item might be “Galvanized reinforcement steel", or “Hot-mix asphalt base course 10in". Item identifiers and descriptions are unique to each state, allowing me high specificity in controlling for project characteristics within states over this decade of data. This project bid-schedule data is the main dataset used for the model estimation.

Additional data I obtain data on firm locations and start years from Dun and Bradstreet, a commercial provider of firm-level data and analytics. I obtain all firms from their database with SIC code 16, “Heavy Construction, Except Building Construction Contractors", and all firms with SIC code 15, “Building Construction - General Contractors and Operative Builders." Appendix B.2 provides details on the matching procedure. Finally, I obtain data from the Census on demographic variables across counties, and from the Bureau of Labor Statistics on price indices.

Summary statistics Table 1 reports summary statistics. Panel A summarizes the data at the auction level, starting with the project shares by the five most common engineering types. Asphalt jobs are the most common, followed by bridges and grading work. The most notable statistic is that the median number of bidders is three, a key fact discussed in Section 3. The variation is large, with a standard deviation of 2. Panel B summarizes the data at the state–year level, showing that the average state has nearly 300 auctions per year, though the distribution is skewed: on average 79 auctions account for 80 percent of annual spending. The average state–year has 145 distinct bidders and 84 distinct winners, a relatively large number given the small average number of bidders per auction.

Panel C reports bidder-level statistics. On average, firms appear for eight years, bid in 51 auctions, and win 14. Firms typically bid in three different types of auctions — for the most part they operate generally instead of being specialized. They also largely face a substantial set of competitors over their lifetime, on average about 32 other firms. Thirty-seven percent never win at all, suggesting

[23] Implemented via OpenAI’s API; see https://platform.openai.com/.

Variable	Statistic	Value
Panel A. Auction statistics		
Engineering Type: Asphalt	Percent	41.7%
Engineering Type: Bridge	Percent	18.9%
Engineering Type: Grading/Excavation	Percent	14%
Engineering Type: Uncategorized	Percent	13.6%
Engineering Type: Concrete Misc.	Percent	2.4%
Tons awarded	Mean	13765
Bidders per auction	Mean	3.63
Bidders per auction	Median	3
Bidders per auction	Std. dev	2.14
Panel B. State statistics		
Auctions per year	Mean	294
Auctions per 80% yearly spend	Mean	79
Bidders per year	Mean	145
Winners per year	Mean	84
Panel C. Firm statistics		
Auctions	Mean	51.37
Wins	Mean	14.12
Years active	Mean	7.54
No. types	Mean	3.01
No. distinct rivals	Mean	32
Never win	Percent	36.9%
Multi-state	Percent	12.4%

Note: Panel A presents summary statistics at the auction level. The five types shown are the five most common auction types. Tons is the total sum of all bid-schedule items that are measured in tons. Panel B averages across each state year. Panel C presents statistics at the firm level. The last two rows give the percent of firms satisfying the variable.

Table 1: Summary Statistics

many test the market but do not stay. However, these firms account for only 2% of all bids, so they are not consequential in the market at large. I identify 12% of firms in the data as operating in multiple states. Because identifiers are at the state level, I conduct this match using both name matching and a large language model; detail and validation results are in Appendix B.2.[24] Among firms that ever win, the interstate share is 18%. This share is low given that, aside from the Rocky Mountains and a few other ranges, there is no natural reason for a state border to segment the market.

[24]My matching approach minimizes false positives and validation implies about a 10% under-count, so a corrected estimate is that 13% of firms are multi-state.

3 Empirical Facts

Infrastructure Prices Keep Rising I begin by examining levels and trends in the key variables of interest, price and competition. Starting with price, I find that over the past two decades, the winning bid for highway infrastructure roughly doubled, as shown in Figure 1. The red line shows the national median winning bid for all roadwork projects, which rose from $1 million in 2002 to $2.1 million in 2024, an increase of 107%.[25] The plot accounts for inflation by dividing by CPI, so this doubling is in real dollars. This pattern extends the increase documented by Brooks and Liscow (2023) for the late twentieth century, and shows continued price growth in the 21st century.

It is possible the rise in winning bids could be caused by a compositional shift towards a different class of projects. This would constitute a rise in expense, but not a true price increase. To examine this, I plot the trend for three additional subsamples of the data, corresponding to more groupings of more homogeneous projects. The orange line shows the trend for asphalt projects, the subset of roadwork in which asphalt work is the primary expense.[26] The yellow line shows the trend for bridge projects, which is a separate category. Both groups have similar increases for winning bids, growing by 134% and 96%, respectively, over the full period. The final line (light blue) turns to project size, plotting the median winning bid for projects between the 45th and 55th percentile of native tons, the best proxy for size available in the data.[27] An advantage of tons is that it captures real differences between projects covering the same length of road with different intensities of reconstruction. A disadvantage is that it only captures inputs measured explicitly in tons. In Appendix Figure A.3 Panel (b) I report a robustness check with hand-collected data for six additional states with a converted tons measure regardless of native units.[28] When controlling for native tons, the price increase is larger, at 150% over the period.

[25] I define roadwork as projects classified as asphalt or concrete pavement, or with descriptions containing specified roadwork keywords (see appendix B.1).

[26] For example, road widening projects in which excavation and earthwork is the primary expense are excluded.

[27] Native tons is the total tons of all inputs that are measured in the unit tons.

[28] The converted tons measure converts all asphalt amounts to tons from their native units. I use standard conversions between volume, weight, and area for asphalt, described in detail in Appendix B.4. Because I only have bid-schedule data starting in 2014, I hand-collect additional data covering the full period for six states. While quite noisy, the measure appears compatible with the price increase.

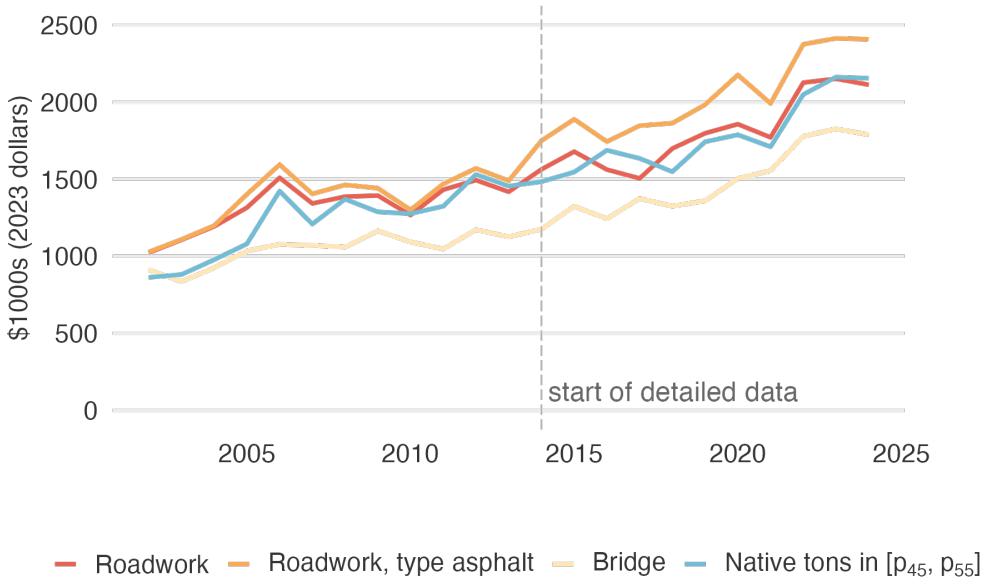


Figure 1: Winning Bids Are Rising Over Time

Note: Figure plots the median winning bid of DOT infrastructure projects between 2002 and 2024. All prices control for inflation and are in 2023 dollars. The blue lines show medians for projects in the 45th to 55th percentile of size, defined as the total tons of input items measured in tons. The percentiles are defined separately for each state. The dashed vertical line marks the beginning of the detailed data, indicating the period covered by the structural estimation later in the paper. Figure A.3 Panel (a) shows analogous trends for a balanced panel of states.

The rise in prices cannot be easily explained. Appendix Figure A.4 Panel (a) shows that the upward trend in prices is just as steep for projects in rural counties and located on flat terrain. Panel (b) shows that the construction PPI has remained flat over most of the period, and average construction wages over the period have not grown at all. Strikingly, the national unionization rate in the construction industry fell by 38% over the period, from 16.7% to 10.3%. Meanwhile, the price of crude oil rose on net since 2002, but fell substantially in 2011, completely at odds with the pattern of winning bids. Crude oil is both the main feedstock of asphalt and the main determinant of fuel and energy prices. Asphalt prices moved with crude oil through about 2010 — nearly doubling and then easing — consistent with the discussion of cost drivers in Mehrotra et al. (2024) and suggesting that asphalt likely contributed early on. Yet a back-of-the-envelope shows only about a 10% impact on total project prices, compared with the 53% increase I observe.[29] It cannot explain the later period, a fact

[29]Calculation uses asphalt at a conservative 30% of pavement materials, materials at about 40% of project cost (Strassner and Moyer 2002, Table 4), and the 80% rise in asphalt prices.

corroborated by the PPI for highway and roads input goods, a new BLS series since 2015, which is very flat, except for a slight post-Covid rise.

Overall the evidence points to substantial real price increases not explained by shifts in project type, location, size, or, for the most part, input costs. I cannot rule out all production-cost drivers or changes in project complexity. Accordingly, in Section 4 I control for detailed project inputs using machine learning and then reverse engineer total production costs from observed bidding behavior, yielding an estimate that captures all cost increases borne by firms.

Price Increases across States are Correlated with Competition Measures I next turn to differences across states. Geography, precedent, and DOT institutional silos produce substantial variation in engineering design across states, confounding simple price comparisons. Unfortunately, engineer terminology in bid-schedules does not overlap across states, preventing the use of these covariates to control for design differences. Appendix Table A.3 illustrates the problem with California and Kentucky. Raw winning bids in California dwarf those in Kentucky by an average of $9.2 million. However, the gap narrows by 82% to $1.7 million after controlling for just a few coarse project design covariates.[30] To reduce the impact of omitted state-level design variables, I rely on the panel aspect of the data and focus on within-state differences. Later, I show that under plausible assumptions, markup level estimates are immune to this bias and can be compared across states.

Table 3 reports correlations between within-state changes in prices and changes in covariates of interest from two-way-fixed-effects regressions.[31] Notably, most of the input price, wage, and demographic covariates have no statistically significant correlation. This includes the coefficient on the state prevailing wage, a policy that has generated discourse (Kessler and Katz, 1999).[32] This is perhaps because the Davis–Bacon Act, the federal prevailing wage law applying to all projects receiving federal funds, is the binding constraint. Regardless, while these production side variables show little correlation, there is a strong positive correlation between within-state increases in price and changes in competition-related measures. States that experienced relatively larger price increases also saw relatively larger declines in the number of active firms and the average number

[30] Under the logic clarified by Oster (2019), instability of coefficients with respect to observables should raise concerns about selection on unobservables.

[31] Appendix Table A.4 reports estimates from parallel regressions that omit state fixed effects. Because project designs differ across states, the resulting correlations are difficult to interpret.

[32] The coefficient is identified by six states that repealed their prevailing wage law over the period.

of bidders per auction, and increases in the Herfindahl-Hirschman Index. These results are purely correlational, yet the magnitudes are non-trivial and motivate further investigation.

	Within-state correlation	
Covariate	Estimated coefficient	(SE)
Log avg. petrol product price	0.147	(0.267)
Log construction wage	-0.014	(0.021)
Pct unionized	0.006*	(0.003)
State prevailing wage law	-0.037	(0.062)
Log population	0.102	(0.126)
Log median HH income	-0.027	(0.025)
Log construction establishments	0.172	(0.258)
Log firms in market	-0.266***	(0.039)
Herfindahl–Hirschman Index (0–1)	0.489***	(0.162)
Log Avg. Bidders	-0.199***	(0.072)
State FE	yes	
Year FE	yes	
Observations (by row): 927, 927, 927, 815, 530, 927, 927, 927, 927, 927		
Adj. R^2 (by row): 0.813, 0.813, 0.814, 0.813, 0.820, 0.813, 0.813, 0.822, 0.815, 0.815		

Note: Table reports estimates from a log-linear two way fixed effect regression of prices (log winning bids) on covariates. The first covariate is the log numer of firms that win any auction in a given year, the second is the Herfindahl-Hirschman Index, and the final is the log number of average bidders per auction. Results are reported the unbalanced and balanced panel.

Table 3: Descriptive Evidence: Price and Competition Indicators

Auctions have Few Bidders I provide new summary statistics on competition in U.S. transportation infrastructure nationally. Figure 2 plots the share of auctions with one, two, or three bidders over time. Overall, most auctions attract few participants. Across the sample, the median is three bidders, and nearly one third of auctions have only one or two bidders. These numbers are more extreme than previous findings in the auction literature; seemingly, the states historically most willing to release data were those with above average competition, perhaps due to general higher administrative capacity.

The time trend in the number of bidders can be broken into two parts. The 2002 to 2009 period reflects the tight link between private sector and public sector construction. As the private construction sector surged with the housing boom, auction participation dropped, driving up the share of low-bidder auctions. With the onset of the Great Recession, this pattern flipped, as stimulus (and lower opportunity costs) boosted public sector construction.

In contrast, the post-2010 period is marked by a secular decline in bidder participation. In 2010, auctions with one to three bidders made up 47% of the total. By 2024, this share had risen to 63%.

Appendix Figure A.5 restricts to roadwork projects and shows a similar increase from a higher baseline: one to three bidder auctions accounted for 59% of all roadwork projects in 2010, and 72% by 2024. The share of roadwork auctions with just a single bidder increased by 40% over the period, from 13% to 18% of all roadwork projects. It is the consequences of this post-recession trend that the model section of this paper tries to understand.

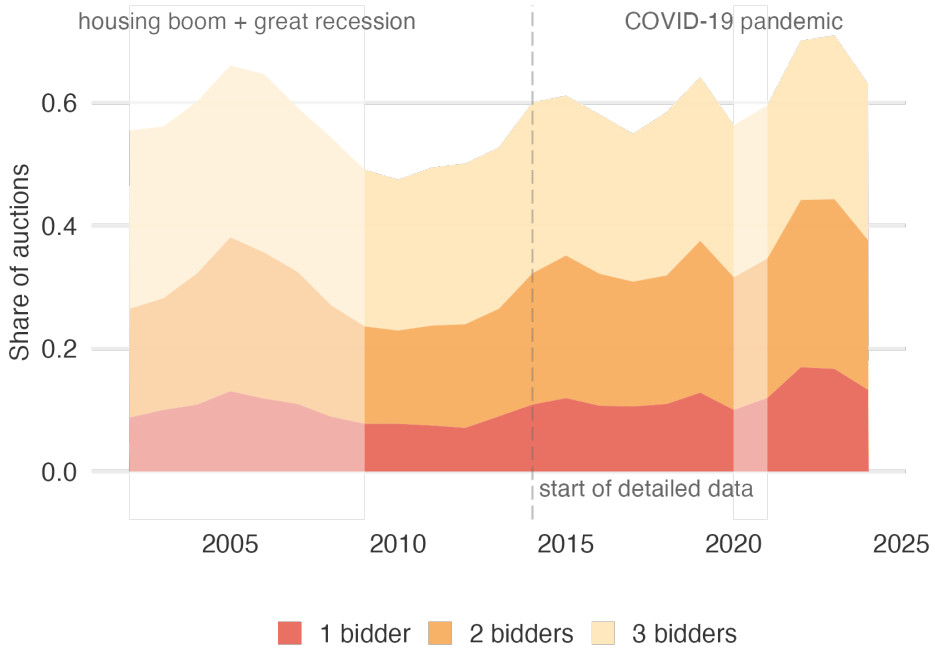


Figure 2: Macro-economic Effects and Decreasing Bidder Trend

Note: Figure plots the share of auctions by number of bidders from 2000 to 2024. The first shaded gray region marks the start of the housing boom and the end of the Great Recession, and the second marks the COVID-19 pandemic. The dashed vertical line marks the beginning of the detailed data, indicating the period covered by the structural estimation later in the paper. Appendix Figure A.5 Panels (a) and (b) plot analogous graphs for the balanced state panel and the roadwork only panel respectively.

One simple mechanism behind the post 2010 fall in bidders could be that infrastructure demand grew faster than firms could enter the market, resulting in fewer bidders per auction. This seems especially possible in the later period given the additional funds allocated to highway infrastructure by the IIJA in 2021. Indeed, entry into the market over the entire period is very limited. As shown in Appendix Table A.2, of all bids in a given year, only 2.5% are placed by a firm new to the market that year. This number increases slightly when also treating firms that cross state lines into new local markets as entrants, but only to 3.3%.[33] However, the story of limited entry and rising demand

[33]The rarity of entry is also inconsistent with a model of low entry barriers and high turnover, as one might expect in contestible market theory (Baumol et al., 1982).

cannot be right, because demand actually *did not* rise over the period. Appendix Figure A.6a shows that the number of projects auctioned by the government has been flat since 2010. Appendix Figure A.6b corroborates this by showing the amount of infrastructure (as measured in total tons) also did not increase over the period, although total spending did.[34] The cyclical trend in annual projects does correlate with the cyclical patterns in bidder participation, but there is no relation in the long run between demand and bidder decline.

First Bid Second Bid Gaps Decline with Bidders One simple indicator of the effect of the number of bidders on price is the gap between the winning bid and the second-lowest bid. This "bid gap" equals the loss to the government if the winning bidder disappeared and all else remained equal. Figure 3 plots the national median bid gap by number of bidders for all roadwork auctions over the sample period. Bid gaps are residualized by state and re-centered at the national average to account for omitted state-level factors that may shift both bidder counts and gap levels.

As the plot shows, the bid gap declines convexly as the number of bidders rises. Since by construction there is no such measure in one-bidder auctions, the largest estimated bid gap is for two-bidder auctions, where the median gap is nearly $300,000 (2023 dollars). The gap falls steeply with additional entrants, leveling off around six bidders at roughly $175,000. Appendix Figure A.7 shows the same pattern appears across different sub-samples, including asphalt projects and projects in the 45th–55th percentile of size.

[34]The lack of an increase in projects built under the Biden administration may be surprising, but infrastructure output during this period remains debated. For discussion, see Zachary Liscow, "Highway investment probably didn't go up under Biden," *Briefing Book*, March 3, 2025.

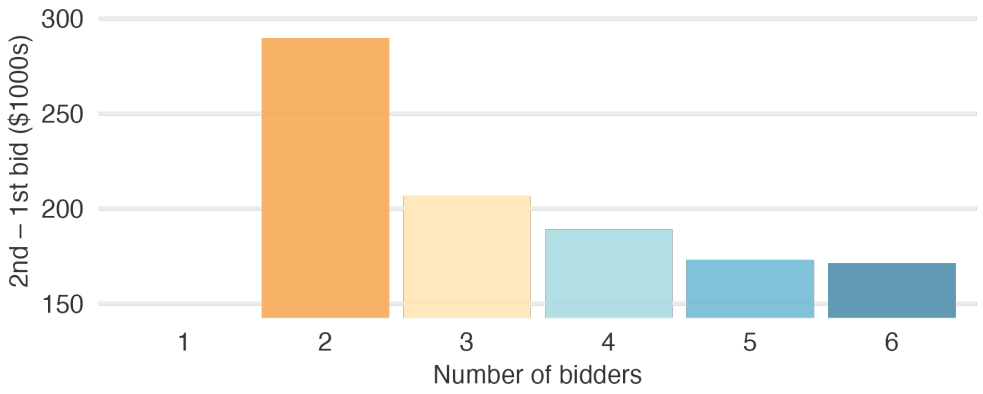


Figure 3: First Price - Second Price Bid Gap by Number of Bidders

Note: Figure plots the national median difference between the second-lowest and lowest bids by number of bidders in the auction for all roadwork auctions. To account for state omitted variables that affect both bid gaps and number of bidders, gaps are demeaned by state with the national mean added back. When taking means, 1% of the sample is trimmed due to large outliers. Appendix Figure A.7 plots analogous bid gaps for three different subcategories of auctions: bridges, asphalt only roadwork, and roadwork projects within the 45th to 55th percentile of native tons, which proxies for a constant size.

Unfortunately, the bid gap is not a measure of the price effect of a new bidder. For it to be so, one would have to make a strong assumption on the composition effect — the new bidder is the low bidder — and the strategy effect — the incumbent bidders do not adjust their bids. While the failure of these assumptions produces oppositely signed biases for the effect of a new bidder, both failures generate a downward bias for the change in the effect of a new bidder.[35] In other words, the decline in the bid gap as the number of bidders increases in Figure 3 is a conservative estimate of the decline in the effect as the baseline number of bidders increases.

Direct Evidence that More Bidders Reduces Prices Exogenous variation in realized bidder participation is rare, and theory itself does not imply that greater potential competition must lead to more aggressive bidding. Li and Zheng (2009) and Jovanovic and Menkveld (2022), for example, develop models in which endogenous participation can offset the usual competition effect. The limited quasi-experimental evidence is also mixed.[36] Motivated by this gap, I directly estimate the

[35] If the first assumption fails, the bid gap overstates the savings to the government, and this failure is more likely when the baseline number of bidders is large. If the second fails, the bid gap understates the savings, and this failure is more likely when the baseline number of bidders is small.

[36] For example, an Italian publicity reform was found to increase bidders by 9.3% and the winning rebate by 7% (Coviello and Mariniello, 2014). However, new e-procurement policies in India and Indonesia were found to increase participation meaningfully (increase of 0.4 bidders from a baseline of 2.9), though not statistically significantly, despite having a precise null (at least greater than -3% in India) effect on contract prices (Lewis-Faupel et al., 2016).

average causal effect of the number of bidders on price, capturing both the composition and strategic effects.

To identify this total effect, I use plausibly exogenous variation in the number of bidders. I would like to estimate

$$\log(\text{price}_{jct}) = \alpha + \beta N_{jct} + \epsilon_{jct}, \tag{1}$$

where $\log(\text{price}_{jct})$ denotes the winning bid in auction j in year t and county c, and N_{jct} is the number of bidders in the auction. The parameter of interest is β, the average causal response to a unit increase in the number of bidders.

The primary challenge in estimating β is that N_{jct} is endogenous, since firms choose to enter auctions based on project characteristics that also affect price. This generates a positive correlation between the number of bidders and the error term, $\text{Cov}(N_{jct}, \epsilon_{jct}) > 0$. The resulting identification problem is formidable, since it requires finding an exogenous shifter of the number of bidders that does not also affect prices. A secondary challenge is that bidders do not observe the number of competitors ex ante. Consequently, I am interested in variation in N_{jct} that also impacts bidders' expectations, as would be consistent with a shift in the equilibrium distribution of N_{jct}, the change of interest. Unpredictable shifts, on the other hand, would likely underestimate the total effect.

To address these concerns, I first summarize at the county level, taking the average of log price, number of bidders, and time-varying covariates for each county-year.[37] County-level changes are more likely to be salient to bidders than auction level variation. I start by estimating the following within-county linear regression:

$$\log(\text{price}_{ct}) = \beta N_{ct} + \lambda x_{ct} + \gamma \tau_t + \mu_c + \epsilon_{ct}, \tag{2}$$

where τ is a time trend and μ_c is a county fixed effect. The variable x_{ct} is average native tons of inputs, my main proxy for size, described in Section 2.2. In order to identify the parameters of

[37] I balance the panel across counties, which drops 36% of observations.

interest, this model needs to satisfy strict exogeneity.[38] While this specification may reduce the bias in equation 1, one is likely concerned that strict exogeneity is violated due to time-varying unobservables that are correlated with both county average price and number of bidders.

For this reason, my preferred approach uses an instrumental-variable (IV), constructed using distance between firms and auctions. Because transportation costs of equipment, materials, and labor are substantial, distance is a strong predictor of firm auction entry. Appendix Figure A.9 reports estimates from a linear probability model of auction entry on distance deciles. At distances under 3 miles, firms enter about 14% of auctions, whereas at 100 miles, the probability falls below 2%. I construct the instrument using firms for whom the given state is a secondary market to ensure their establishment location is exogenous to the affected auctions. Appendix Figure A.8 visualizes the variation used in the instrument, showing a single firm's expansion over time. The expansion across the border is notably lumpy in time and the auctions entered are driven by distance, suggesting this instrument has a strong first stage.

Specifically, to construct the instrument, I identify each firm i's home state h_i as the first state they enter. For state s and year t, I define an out-of-state bidder as a firm with at least one bid in s during t whose home state is not s, i.e. if $|\text{bid}_{ist}| > 0, h_i \neq s$. I calculate the distance between i's location and the county of every auction in s during t. Details on the construction of the distance measure are in Appendix Section B.5. I construct a county level indicator for whether there is an out-of-state bidder within 100 miles:

$$N_{ct} = \gamma \mathbb{1}(\min_i(d_{ict}) < 100 \text{ miles}) + x_{ct} + \varepsilon_{ct}. \tag{3}$$

I choose 100 miles based on the entry and distance regression reported in Figure A.9, but report results with varying this threshold in Appendix Table A.5. For the instrument to be valid, I require (i) distance strongly predicts auction entry (relevance); (ii) firm locations are as-good-as random

[38] Strict exogeneity requires that, conditional on the county fixed effect, the entire path of the error is independent of the entire path of the regressors: $(\epsilon_{c1}, \ldots, \epsilon_{cT}) \perp\!\!\!\perp (Z_{c1}, \ldots, Z_{cT}) \mid \mu_c$, with $Z_{ct} = (N_{ct}, x_{ct}, \tau_t)$. I also present a similar exercise using a first-difference specification in the Appendix Table A.5. Under first-differences, unbiased estimation of β requires that $\Delta\epsilon_{c,t}$ is uncorrelated with the contemporaneous change in the independent regressors.

with respect to cross-border project proximity (random assignment); (iii) out-of-state firms affect prices only through the number of bidders (exclusion), and (iv) the instrument never reduces the number of bidders per auction (monotonicity).[39] I argue the second assumption is reasonable based on the observation that a firm's home state is typically its primary market (88% operate exclusively in one state). For firms that enter markets outside their home state, the average time until entry is 6 years. The geography of out-of-state auction opportunities is therefore less likely to impact firm location choices than in-state auction opportunities. Should this assumption still fail, the selection bias would shrink the magnitude of the estimated coefficient.[40] A potential threat to assumption (iii) would be if out-of-state bidders have systematically lower bids than state incumbents. Appendix Figure A.10 shows that out-of-state bidders actually have a slightly lower probability of winning an auction, suggesting higher bids, but the confidence intervals between out-of-state and control bidders overlap.

Table 4 reports the estimated coefficients. Column (1) shows the ordinary least squares result yields a positive correlation between the number of bidders and price, consistent with selection bias. Column (2) shows that county fixed effects alone reduce the selection bias and flip the sign of the point estimate. Column (3) reports the first stage for the IV regression, showing that proximity to out-of-state bidding firms significantly increases the number of bidders in an auction. Switching from no out-of-state bidders within 100 miles to at least one raises the number of bidders on average by .15.

[39] I follow the terminology of Imbens (2014).

[40] An alternative reason for estimates to be biased toward zero is if panel transformations diminish the fixed-effect signal relative to measurement error, as emphasized by Griliches and Mairesse (1995).

	Dependent variable:					
	Log Bid	Log Bid	No. bidders	Log Bid	Log Bid	Log Bid
		(Within County)	(First Stage)	(IV)	(IV)	(IV)
No. bidders	0.071***	−0.014***		−0.125**	−0.097*	−0.152
	(0.004)	(0.003)		(0.056)	(0.050)	(0.098)
dist. firm < 100mi			0.153***			
			(0.018)			
Log tons	0.096***	0.149***	−0.017***	0.135***	0.126***	0.135***
	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)
State FE			yes	yes	yes	yes
Time Trend		yes	yes	yes	yes	
County FE		yes				
Year FE						yes
Lag log bid		yes				
First stage partial F				76.26	102.79	27.48
Observations	22,050	22,050	22,050	22,050	13,461	22,050
Adjusted R^2	0.125	0.562	0.303	0.471	0.458	0.459
Note:						*p<0.1; **p<0.05; ***p<0.01

Note: Table reports coefficient estimates from equation 1 using the strategies outlined in equations 2 and 3. Column (2) includes a lagged outcome to account for dynamic bias in fixed effect panel models, which Klosin (2024) shows is greater than Nickell bias. Robustness checks to the distance threshold in the instrument are in Appendix Table A.5.

Table 4: Estimates of the Impact of Number of Bidders on Prices

Columns (4) - (6) present results from the instrument specification. All three specifications produce estimates of a similar and economically meaningful magnitude. Column (4) directly follows the specification in equation 3, while column (5) restricts the sample to counties within 50 miles of a state border and, reassuringly, estimates a similar, though slightly smaller effect, suggesting that the average causal effect of one more bidder is a roughly 10% decrease in price. To be conservative, this is my preferred estimate. Finally, column (6) replaces the time trend with year fixed effects; the estimate is not statistically significant, but the point estimate is similar and slightly larger in magnitude. Appendix Table A.5, Columns (2) to (5), vary the distance thresholds. While the smallest cutoff reduces statistical significance, overall the estimates remain consistent with an effect of approximately 10%.

These results say that, for the set of U.S. auctions whose number of bidders are affected by out-of-state entrants, on average one additional bidder reduced the government's payment by 10%. This leaves at least three questions open: (i) how do effects vary, in particular with the baseline number of bidders; (ii) can a decline in competition explain the path of rising costs over time; and (iii) what are the

prospective savings if we were able to increase bidder participation in the future. To address these, I develop a model below.

4 Auction Model

I present a model of bidding in procurement auctions to decompose bids into production costs and markups. The model builds on an independent private values auction model, allows the government to reject bids through a secret reserve price, and addresses bidders' uncertainty about competitors with a probability distribution over the number of bidders. Given this setup, the structure imposed by the auction format determines firms' optimal actions, identifying the production cost with minimal functional-form assumptions, as shown in Guerre et al., 2000. The richness of my auction data enables me to preserve this functional form flexibility, using machine learning methods to control for project heterogeneity. I estimate a separate auction model by state, year, and clusters of similar auctions, in order to fully capture patterns across states and over time.

4.1 Setup

I consider the problem faced by a firm bidding in an auction to win a contract to construct a road. Each project is heterogeneous and distinguished by project-specific covariates X. I omit auction subscripts to simplify notation; however, covariates, costs, and bids are always auction-specific. A firm's optimal bidding strategy trades off raising margins above its cost of constructing the road with its chances of winning the auction. I assume firms bid competitively; Appendix Section C.1 discusses collusion and presents both descriptive evidence and formal tests showing that collusion is unlikely to be pervasive, though it cannot be ruled out in principle.

I make the following assumptions.

(A1) *Costs.* Bidder i has a cost c_i of completing the project drawn independently from a distribution with support $[\underline{c}, \bar{c}]$, and c.d.f. $F(\cdot \mid X)$. I assume $F(\cdot \mid X)$ is twice continuously differentiable on $[\underline{c}, \bar{c}]$ and its derivative is strictly positive on $[\underline{c}, \bar{c}]$, and $f(\bar{c}|X) > 0$.

(A2) *Information I.* Bidder i does not know the total number of bidders N, but the probability

distribution of N, $p_{N|X}(n) \equiv \Pr(N = n \mid X)$ for $n \in \{1, \ldots, \bar{n}\}$ is common knowledge.

(A3) *Information II.* Bidder i knows her own cost c_i, does not know her competitors' costs $c_{j \neq i}$, but the distribution of all costs $F(\cdot \mid X)$ is common knowledge.[41]

(A4) *Risk neutrality*. Bidders are risk neutral and therefore maximize expected profit.

(A5) *Government reserve price.* The government sets a reserve price r for each auction such that bids exceeding r are rejected. The reserve price is secret, meaning it is unknown to bidders. Reserve r is independently drawn from the distribution $H(\cdot|X)$, where the distribution is common knowledge. Conditional on X, r is drawn independently from c_i.

Assumption (A1) rules out a common-value setting in which firms receive noisy signals about a shared underlying cost.[42] Such an environment may generate the winner's curse, where the winning firm systematically underestimates costs.[43] As discussed in Section 2.2, empirical estimates of post-auction renegotiation are modest, with the median ranging from -4% to 3% for four sampled states. While not a test, if the winner's curse were prevalent, one might expect substantial renegotiation to cover losses.

Assumptions (A2) and (A3) together reflect the reality that bidders have limited information on the competitors they face. While bidders may be informed about the pool of potential competitors, it is effectively impossible to know how many will ultimately bid. According to a survey by Liu et al. (2022), most DOTs publicly disclose no information on approved bidders before auctions. For the remainder that do, these lists are often large and bidders can at most form a prediction. As shown in Table 1, on average there are 145 unique bidders and 84 unique winners per state per year. Firms do not partition neatly across sub-markets, as the majority of firms bid on multiple project types and

[41] It would be possible to add firm heterogeneity in the cost distribution. It does not matter in this case, as firms do not know who their competitors are when bidding, so their strategy is only a function of the distribution over all possible firm types.

[42] A1 also assumes finite (bounded) support and positive density at the upper endpoint $\bar{c}$, which provide the initial point needed to apply the fundamental theorem of ODEs to prove a unique equilibrium bid function, as in Maskin and Riley (2000). I conjecture that the argument here follows by adapting their Proposition 2.

[43] Alternatively, it may induce bidders to shade their bids conservatively, as Hendricks et al. (2003) provide evidence of in offshore oil lease auctions. In that setting, the value of a lease depends on the uncertain size of subsurface oil reserves. In contrast, roadwork projects have far fewer unknowns, as they are standardized and have detailed specifications provided in advance.

in multiple locations: the average firm bids in 11 counties; among firms with at least 10 bids, the average is 25 counties.

Assumption (A4) assumes firms maximize their expected profit. Assumption (A5) models the government's rejection of bids judged "unreasonable," i.e., bids that fail screening heuristics. Most often, these rules incorporate a percentage markup over the DOT's engineering estimate, along with other factors (see Liu et al. (2022)). Other papers have differed in their approach to modeling this feature, often treating the government as an additional bidder, or ignoring it entirely.[44] As more than 10% of auctions in my data have a single bidder, it is necessary to directly account for the threat of rejection.

I model the government as having a secret reserve price r with distribution $H(r) = F_b(r/1.15)$, where F_b is the distribution of bids. This assumption mimics the common screening threshold of 1.15 times the engineer's estimate. Appendix Figure A.11 shows that, across the 18 states with engineer's estimates in the data, the bid distribution closely matches the distribution of engineers' estimates. This is not surprising, as past bids are the main information engineers have with which to form their estimation. While imperfect, this assumption offers a parsimonious representation of a partly qualitative process and avoids modeling the government as a strategic bidder. It is conservative: in practice, DOTs are often hesitant to reject bids, so any bias in markup estimates is likely to be downward.

Each firm i in the auction submits a bid b_i that maximizes its expected profit. Firm i's profit conditional on winning is simply the difference between the bid and their production cost. Firm i's expected profit prior to the outcome of the auction is

$$\mathbb{E}\pi_i = (b_i - c_i)\Pr(b_i : b_i < \min\left(\left\{b_j : j \neq i\right\}, r\right) \mid X). \tag{4}$$

While in general no closed form solution exists for a standard first-price auction with a secret reserve price, there exists a Nash equilibrium bidding strategy β, which is symmetric and increasing in costs.[45] Consequently, the profit condition for bidder i can be written as

[44] For example, Li and Zheng (2009) , Bajari, Houghton, et al. (2014), and Krasnokutskaya and Seim (2011) assume no reservation price for Texas and California DOT auctions, while Bhattacharya et al. (2014) assume a publicly observed reserve price set at 1.5 times the engineer's estimate for Texas auctions.

[45] I assume this is the unique equilibrium, following Maskin and Riley (2000), which provides the uniqueness proof

$$\mathbb{E}\pi_i = (b_i - c_i) \sum_n \left[\Pr(N = n \mid X) \left(1 - F\left(\beta^{-1}(b_i)|X\right)\right)^{n-1} \right] \cdot (1 - H(b_i|X)),$$

relying on the symmetric increasing bid strategy to replace the probability firm i wins with the probability all firms other than i draw costs higher than i's cost, and to write the inverse bid function. The first order condition with respect to the bid defines the bidding strategy as a differential equation:

$$\beta(c_i) = \underbrace{c_i}_{\text{variable cost}} \underbrace{- \left. \frac{\Pr_{\text{win}}(b_i)}{\Pr'_{\text{win}}(b_i)} \right|_{b_i=\beta(c_i)}}_{\text{firm margin}}. \tag{5}$$

The derivation is straightforward and is in Appendix D.1 with the explicit equation. Here and throughout the paper, I define *the margin* as the difference between the bid and the marginal cost and *the markup* as the margin over cost. The margin decreases as the probability a high number of firms enter the auction grows. The amount it decreases is mediated by the underlying distribution of costs. To understand the importance of the cost distribution intuitively, consider a degenerate distribution where all firms have the same cost. In that case, the equilibrium strategy is to bid one's true cost — the margin is 0 if there are at least two bidders. On the other hand, if there is variation in cost draws, an additional bidder increases the magnitude of $\Pr'_{\text{win}}(b_i)$, pushing down the margin.

4.2 Identification and Parameterization

Semi-parametric Identification

My identification argument closely follows Guerre et al. (2000). Let $F_b(\cdot)$ denote the distribution of bids. The unique equilibrium strategy means that $F_b(b_i) = F(\beta^{-1}(b_i))$, using the equality $b_i = \beta(c_i)$. The density of bids is therefore $f_b(b_i) = f(\beta^{-1}(b_i))/\beta'(\beta^{-1}(b_i))$. Substituting these terms into the bidding equation allows for an expression in terms of observables:

for a similar class of auctions.

$$c_i = b_i \; - \; \frac{S(b_i)\left(1 - H(b_i)\right)}{S(b_i)\, h(b_i) \; - \; S'(b_i)\left(1 - H(b_i)\right)}, \tag{6}$$

where $1 - H(b_i)$ is the reserve component, $S(b_i) \equiv \sum_n \Pr(N = n)\, A(b_i)^{n-1}$ is the rival response term, and $A(b_i) \equiv 1 - F_b\big(b_i \mid X, p_{N|X}\big)$ is the bid-distribution term. The bid distribution F_b is conditional on $p_{N|X}(n)$, the probability mass function for the number of bidders, because strategies are conditional on $p_{N|X}(n)$. For comparison, the standard model that assumes the number of bidders N is known, requires that F_b is conditional on N; the analogue here is more complicated since any shift in $p_{N|X}(n)$ changes the bid strategy.

Since bids are observed and I have assumed H is a scaled distribution of bids, observation of $p_{N|X}(n)$ renders the right-hand side of equation 6 observed, identifying costs. In practice, I will group auctions into two clusters, a “many-competitor" and a “few-competitor" group, using machine learning and their bid-schedule characteristics. I will estimate $p_{N|X}$ as the empirical distribution of N within its cluster.

Semi-parametrization Equation 6 contains high-dimensional conditional densities, the estimation of which is complicated by the curse of dimensionality. To circumnavigate this issue, I modify the approach of Haile et al. (2003) to essentially residualize out project-specific covariates from bids. My approach takes advantage of my rich covariates, machine learning, and the fact that I observe *all* bids, not just winning bids. The necessary theoretical assumption for the residualization is that costs are multiplicatively separable into two components: an auction-specific term common to all bidders, and a bidder-specific idiosyncratic term. Specifically, I assume

$$c_{ia} = \exp\left(X_a'\gamma\right)\mu_a \tilde{c}_{ia};\; (X_a,\; \mu_a,\; \tilde{c}_{ia}) \text{ are mutually independent,}$$

where μ_a is an auction-specific cost shifter. The benefit of this assumption is that it directly implies

$$b_{ia} = \exp\left(X_a'\gamma\right)\mu_a \tilde{b}_{ia}, \tag{7}$$

where $\tilde{b}_{ia}$ is the bid corresponding to an auction with $c_{ia} = \tilde{c}_{ia}$, i.e. the cost for bidder i in a hypothetical auction with covariates $\exp\left(X_a'\gamma\right)\mu_a = 1$. The proof is in Appendix D.2. Note that I have already assumed the government reserve follows a scaled distribution of the bid; for statement 7 to hold, I require the less restrictive assumption that whatever distribution r follows, it scales with the same $\exp(X_a'\gamma)\mu_a$ factor. This assumption, of course, is still restrictive in that it assumes the government's reserve fully adjusts in response to project cost shifters. While this is what most DOTs attempt to do, they may suffer from incomplete information.

The fact that bids are homogeneous of degree one in costs means that $\tilde{b}_{it}$ from regression (7) can be interpreted as bids for hypothetical homogeneous projects. This individual component of the bid is thus comparable across auctions and suitable for estimating the distributions and densities in equation (6). The resulting estimator is semi-parametric: I have assumed that the distribution of bids belongs to a scale family, where the scale parameter depends on covariates.[46] Estimated margins for the homogenized projects can be scaled back to true bid magnitudes by respectively multiplying each by $\exp(X_a'\gamma)\mu_a$. A useful implication is that the markup for the homogenized auction is the same as for the original auction.

4.3 Estimation

Step 1: Bid homogenization I estimate $\ln \tilde{b}_{ia}$ as the residuals from the regression

$$\ln b_{ia} = X_a'\beta + \alpha_a + \epsilon_{ia}, \tag{8}$$

where X_a as a vector of project covariates, including (i) fixed effects for the project engineering type project and the AI-derived project classification; (ii) location ruggedness and rurality measures; (iii) the logarithm of the total quantities of material, summed separately by measurement unit: tons, cubic yards, linear feet, square yards and counts of "each"; and (iv) the logarithm of the number of distinct material items listed in the bid schedule. To avoid overfitting given the large number of covariates, I regularize the project coefficients using a ridge penalty.

The variable α_a indicates a project random effect, which I can estimate because the data includes all

[46] Note that the probability distribution of N does not need to be homogeneous across projects for estimation.

bids per auction, not just the winning bid. The random effect models any remaining unobserved heterogeneity across projects. I use the equivalence between regularization methods and hierarchical bayesian methods to estimate regression 8 with maximum likelihood and estimate the coefficients and the variance of α_a.[47] Estimation is done separately for each state, allowing the covariate–cost relationship to be state-specific. I include year fixed effects but do not residualize by them when solving for ϵ_{ia}.[48] The prediction $X_a'\hat{\beta} + \hat{\alpha}_a$ therefore accounts for changes in cost driven by changes in covariates, while ϵ_{ia} is an estimate of the cost for a fixed project, which make shift over time. I normalize every homogenized bid to a benchmark asphalt resurfacing and repair project—at national median ruggedness and rurality—using each state's predicted material quantities for the national median asphalt project size (Appendix B.4). However, this normalization is not important since the results that follow mostly do not rely on comparisons of cost levels across states.

Step 2: Cluster auctions To estimate $p_{N|X}(n)$, I cluster auctions based on their covariates, state, and year, and then use the empirical distribution of N within cluster. In practice, I use two covariate-based clusters, which I term the "many-competitor" and the "few-competitor" type. To assign projects, I apply a gradient tree boosting algorithm to predict the number of bidders using the full set of bid-schedule items and split projects at the median.[49] Recall that the goal of $\widehat{p_{N|X}(n)}$ is to reflect bidder beliefs, not to perfectly predict N. Although beliefs are unobserved, Appendix Table A.16, columns (4) and (6), shows the predicted number of bidders correlates more strongly with homogenized bids than the actual number of bidders does, consistent with the prediction aligning more closely with bidders' expectations. Figure A.15 shows the distribution of the actual number of bidders across the clusters; the distributions are distinct but overlap in the right tail of the "few-competitor" group and the left tail of the "many-competitor" group.

Step 3: Bid inversion I recover costs by applying a nonparametric bid inversion using equation 6, in the spirit of Guerre et al. (2000). Within each state, year, and auction cluster, I estimate the bid

[47] An L2 (ridge) penalty on a coefficient block is equivalent to modeling that blocks as random effects if the Langange multiplier satisfies $\lambda = \frac{\sigma_\epsilon^2}{\tau^2}$ where σ_ϵ^2 is the error variance and τ^2 is the variance of the prior. See Hastie et al. (2009) for details.

[48] The cost distribution need not be homogenized over time for auction estimation, because it done separately by year.

[49] The advantage of using a machine learning algorithm here, relative to bid homogenization, is threefold. First, since I no longer have multiple observations per auction, I cannot estimate auction-specific fixed effects. Second, the comparability of covariates across states is no longer necessary. Third, the relationship between auction entry decisions and inputs is likely more complex than the relationship between construction costs and inputs.

distribution and density implied by $\tilde{b}_{ia}$ and substitute these objects into the equation. To estimate the bid density, I follow Hickman and Hubbard (2015) and apply a boundary bias correction to the kernel density estimator, which improves performance near the support of bids and avoids the need for sample trimming. To obtain actual rather than homogenized costs, I rescale the results, reversing the homogenization procedure above.

Robustness Appendix Table A.15 and Figure A.13 present a summary table and accompanying violin plots for regression 8 and two alternative specifications for bid homogenization. The first alternative is a parsimonious baseline that retains only the project classification, engineering type, county ruggedness and rurality, and total tons. The second augments the baseline with exact quantity measures for the individual bid-schedule items, conditional on items having passing a prespecified threshold frequency; for example, "wet-weather pavement-marking tape" is included when it is sufficiently common. Because the resulting set of items is extensive, I estimate these regressors under a moderate ridge penalty. Full estimation details are provided in the appendix.

The table shows that the preferred specification achieves the lowest cross-validated root-mean-squared error on every reported metric, and the violin plots confirm that, while the three residual distributions are broadly similar, the preferred model yields the most concentrated distribution.

Appendix Table A.16 compares predictions of the number of bidders from the gradient tree boosting algorithm described above to those from a penalized multinomial logit model. Columns (1) and (2) show that gradient boosting achieves substantially higher R^2 values in explaining both the actual number of bidders and the resulting bids. Figure A.14 visualizes the bidder count predictions.

5 Estimates

5.1 Estimates of Costs and Markups

Table 5 reports summary statistics of estimates for costs and margins over roadwork auctions in the continental U.S. from 2014 to 2023. Reported costs and margins are rescaled back from the homogenized bids to match the actual bid levels, reversing the homogenization procedure in Section 4.3. Although estimation uses all bids, I report results across winning bids only to represent realized

prices. The first panel reports auction-level summaries over all winning bids; the second panel reports the interquartile range across state means; the final panel groups projects by terciles of estimated costs and reports results by tercile.

Estimated margins are economically large. The mean margin is approximately $0.7 million against a mean cost of $2.5 million. Correspondingly, the mean markup over all winning bids is .22, meaning for the average project the government pays 22% above the production cost.[50] There is a wide range in the estimated markup across projects, with the 25th percentile markup equal to .1 and the 75th percentile triple that, at .34. The median markup is slightly lower than the mean, at .17, reflecting a long right tail. To address concerns that extreme tails are due to estimation error, all means are estimated with a 1% sample trim; Appendix A.6 shows that the mean markup is stable under larger sample trims.[51]

[50] Estimated markups in procurement auctions vary widely. For example, Krasnokutskaya (2011) estimates markups of about 8% in Michigan, while Li and Zheng (2009) finds profits of roughly 26% of bids in Texas. In the broader literature, this estimate is close to the benchmark in De Loecker et al. (2020), who report a median markup of about 20% for U.S. public firms.

[51] A small number (less than 3%) of estimated costs are negative. These arise when extremely low bids can only be rationalized by negative costs, which may reflect genuinely negative costs (e.g. if a firm needs to use sitting materials) or dynamic incentives to win, violations of the model such as non-risk-neutrality, or failure to control for some project heterogeneity. This project pushes hardest on the third concern using rich controls, machine learning, and random effects, though bias may persist.

Variable	Mean	Median	25th pctl	75th pctl
Auction-level				
Markup	0.22	0.17	0.10	0.34
Cost	2477	1251	401	3210
Margin	693	339	147	795
State-level (means)				
Markup			0.19	0.26
Cost			2097	4100
Margin			588	879
By project cost				
Small				
Markup	0.62	0.28	0.15	0.67
Cost	436	416	259	608
Margin	222	118	64	257
Medium				
Markup	0.26	0.17	0.11	0.30
Cost	1606	1537	1158	2028
Margin	405	268	167	474
Large				
Markup	0.18	0.13	0.09	0.20
Cost	7013	5129	3608	8431
Margin	1161	784	467	1436

Note: Table reports the mean and interquartile range of winning bids for roadwork projects from 2014–2023. All monetary values are in thousands of 2023 USD. The bottom and top 1% of observations are trimmed to remove extreme tails. Appendix Table A.6 reports means under alternative trim percents of 0%, 1%, 3%, and 5% trim; the markup stabilizes after the 1%. For the third panel, only projects with estimated winning costs greater than $.1 million are used, dropping the smallest 10% of auctions.

Table 5: Summary Statistics of Estimates

Some state procurement markets appear noticeably more compelling to bidders. Variation in the markup across states is substantial. The 25th-percentile state has a mean markup of 0.19, whereas the 75th-percentile state has a mean of 0.26. The second row in this panel shows that costs also differ markedly across states, nearly doubling from the 25th to 75th percentile state. This variation reflects a combination of differences in project design and differences in the cost of a given project holding design fixed. Unfortunately, because detailed project covariates are defined *within* state, distinguishing between these two channels is complex. However, as discussed in Section 4.2, under the assumption of multiplicative project covariates, the markup is scale-invariant and therefore directly comparable across states, regardless of the baseline cost differences.

High-markup states are concentrated in the Midwest and the South, as shown in the map in Appendix Figure A.17. Kentucky stands out with a median estimated markup of 0.55, reflecting a remarkably low average number of bidders per auction (1.6 in the estimation sample). Large markets such as

California and Texas exhibit lower markups. Rhode Island is an outlier, achieving low markups despite its small market size. However, this likely reflects unusually high cross-border participation, since the state has the highest share of multistate firms active within its borders (74% over the period).

The third panel of Table 5 shows that markups decline with project size. The median falls from 0.28 in the smallest tercile to 0.13 in the largest. In auctions, markups adjust with both the distribution of costs and the number of bidders. In the setting at hand, an additive increase in costs (a location shift of the cost distribution) that leaves its shape and scale unchanged keeps the margin constant and lowers the markup. By contrast, if costs increase with a multiplicative scaling of the cost distribution, the margin increases while the markup is constant. Empirically, the dispersion of both costs and prices appears to increase as mean increases, which would suggest larger projects see higher margins and variable profits, and even higher markups if the effect is large enough. At the same time, bidder counts are not fixed, and in fact increase with project size from 2.69 to 3.18, which in turn pushes markups down. On the whole, the net effect of these two forces results in declining markups. The model so far is agnostic about why certain auctions draw a higher number of bidders; Section 7 takes up that question.

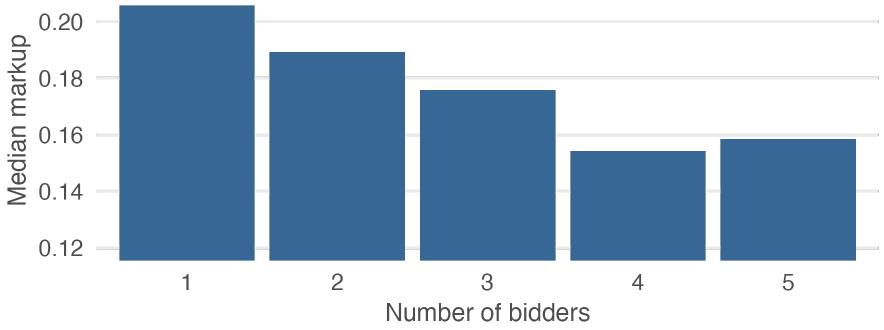


Figure 4: Estimated Markups by Realized Number of Bidders

Note: Figure plots the median estimated markup by actual number of bidders across all winning bids for roadwork projects from 2014-2023. Markups are computed from estimated costs and margins; observations with non-positive costs/markups are excluded. Bars reflect unweighted medians across auctions (not value-weighted).

Figure 4 plots estimated markups against the realized number of bidders, paralleling the bid gap curve in Figure 3. The median markup drops by about one quarter from one to four bidders. Because bidders observe only the distribution of bidder counts rather than the realization, the pattern is

generated by between state–year–cluster variation. The figure reports up to five bidders; beyond this point, most auctions are classified into the high-competitor cluster, mechanically restricting the incremental effects.[52]

5.2 Decomposing price growth over time

Using the estimates of costs and markups for every auction between 2014 and 2023, I decompose the price trend over this period into its component parts. Because estimation is split by year, the time trends in estimated costs and margins are allowed to be fully flexible. I first compare the actual mean price with the mean price for homogenized projects, which control for detailed project characteristics as discussed in Section 4.3. The gap captures changes in observed project characteristics—including shifts in project type, size, and input mix—which I refer to broadly as project complexity. Changes in input prices, however, are not removed by this homogenization and instead appear in estimated production costs. The mean estimated production cost and mean estimated margin together sum to the mean homogenized bid each year.

Figure 5 visualizes the decomposition. The red dashed line shows the mean actual bid, which rose 28% over the ten-year period in real terms. The yellow line shows the mean homogenized bid, which rose 18%. Thus, changes in observed project characteristics account for 36%, or roughly one third, of total price growth. Appendix Figure A.16 illustrates what these changes can look like in Nebraska: growing inputs include environmental adjustments, technical items such as membrane waterproofing, and training, while the average number of inputs per project increased by 19%, from 54 to 64. These patterns suggest that projects are becoming more complex, requiring both more inputs and, in some cases, more technically demanding ones.

Even after holding observed project characteristics fixed, two-thirds of the price increase remains. The shaded regions in Figure 5 decompose this remaining growth into production costs and margins. Over the full period, estimated production costs rose only 8%, while estimated margins rose 44%, with margin growth accounting for roughly two thirds of the remaining real price increase. The limited growth in production costs is notable because the cost estimated in the model represents

[52]The slight uptick at five bidders may reflect coarse clustering; finer clusters trades off increasing the similarity of projects estimated together with a loss in precision. The upward turn at five bidders could also arise from genuine variation, e.g. if five-bidder auctions have higher cost variance which generates larger markups.

all variable costs of production, which together form the basis of a firm's bid. This includes labor, physical inputs, permitting and delay costs, and the firm's opportunity cost. Individual components may have moved differently over time, but aggregate production costs rose relatively little. The decomposition therefore points to rising margins, rather than rising production costs, as the primary source of the increase in the price of a comparable project.

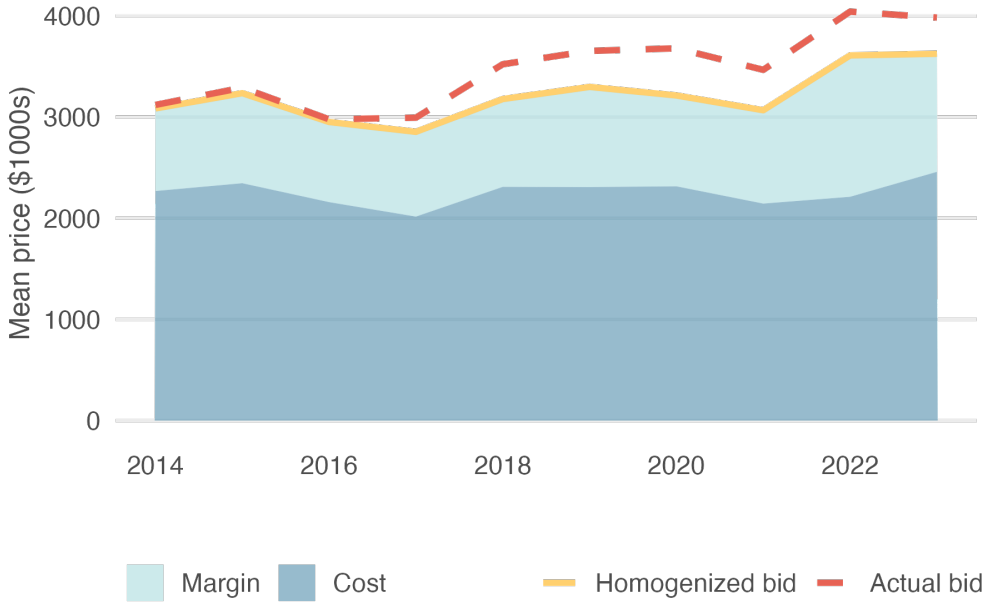


Figure 5: Decomposition of Price over Time

Note: Figure decomposes the average winning bid between 2014 and 2023 into changes due to project characteristics, estimated production costs, and estimated margins. The red dashed line shows the mean actual bid, while the yellow line shows the mean homogenized bid. Both series are expressed in real 2023 dollars and exclude the top and bottom 1% of bids. The dark shared region represents the production cost and the light shaded region the markup.

6 Counterfactuals

How much would the government save if bidders faced more competition? I estimate the prices the government would pay if each auction had one additional bidder. Unlike the previous reduced-form exercise, the model allows me to account for the current distribution of bidders. As Figure 4 demonstrates, the effect of one more bidder depends on the current baseline. Holding bidders' cost fixed, I estimate counterfactual bidding strategies under the hypothetical guarantee of one

additional bidder in every auction. Specifically, I shift the bidder-count distribution up by one – i.e., the probability mass on N bidders is reassigned to $N+1$. The model makes explicit how bidders' beliefs and strategies adjust, allowing for the decomposition of the total price effect into the bidders' strategic response and the composition effect generated by the expanded bidder pool.

Estimation Having estimated costs under the observed distribution of bidders, $p_{N|X}$, I now hold these costs fixed and derive counterfactual bids under an alternative bidder distribution, $p^{cf}_{N|X}$. The bidding strategy comes from equation 5, restated fully here:

$$\beta = c_i + \frac{\sum_n \Pr(N=n|X)\left(1-F(\beta^{-1})\right)^{n-1}\left(1-H(\beta)\right)}{\sum_n \Pr(N=n|X)\left(1-F(\beta^{-1})\right)^{n-2}\left((n-1)\frac{f(\beta^{-1})}{\beta'}(1-H(\beta))+\left(1-F(\beta^{-1})\right)h(\beta)\right)},$$

where I have omitted conditioning covariates for legibility. As before, β is the bid strategy, c_i is bidder i's random cost draw, F is the distribution of costs, H is the distribution of the secret reserve price, and $p_{N|X}$ is the probability distribution of the number of bidders.

The secret government reserve price complicates estimation relative to other empirical auction settings. If there were no secret reserve price, the equilibrium bid solves a straightforward ordinary differential equation with solution

$$\beta = \mathbb{E}\left[\min\left\{C_j : j \neq i\right\} \mid \min\left\{C_j : j \neq i\right\} > c_i\right] = \frac{1}{G(c_i)}\int_{c_i}^{\bar{c}} t g(t) dt,$$

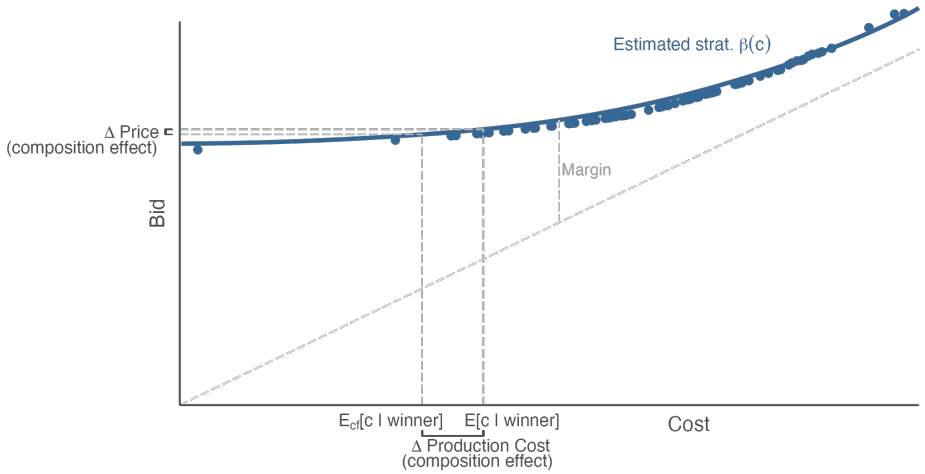
where $G(c_i) \equiv \sum_{n=1}^{N} \Pr(N=n)(1-F(c_i))^{n-1}$. Under this bid strategy, counterfactuals could be computed straightforwardly by integrating over the mixture distribution $G(c_i)$.

With the secret reserve price, the ordinary differential equation cannot be solved analytically. Therefore, I instead estimate the bid function numerically from an initial boundary condition. Intuitively, the solver expresses β' directly as a function of β, iteratively tracing out the equilibrium bid curve from this initial point. The estimating equation and implementation details given in Appendix D.3.

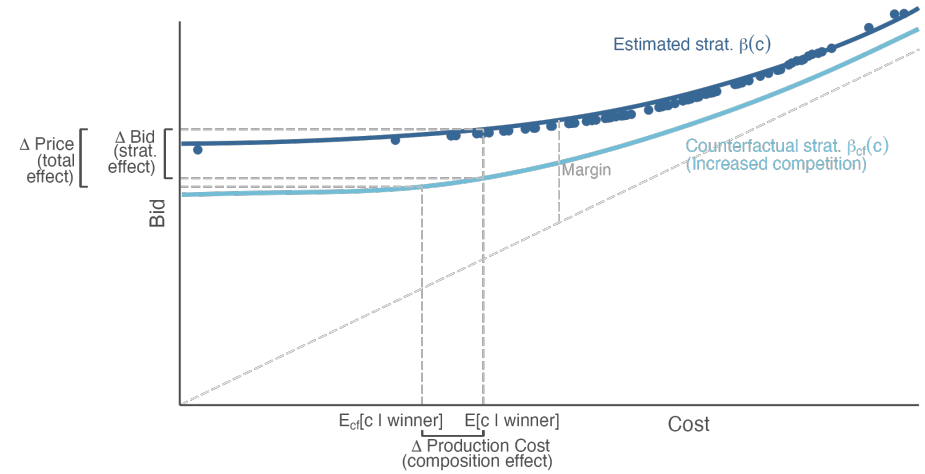
Results An additional bidder lowers project prices in two ways. First, by placing additional bid pressure on incumbents, and second, by sometimes having the lowest cost draw. Figure 6 illustrates these two mechanisms with real data for one example state. The shift from the estimated strategy curve $\beta(c)$ to counterfactual strategy curve $\beta_{cf}(c)$ reflects more aggressive bidding – for a given cost draw, a firm now places a lower bid. This is the strategic effect. The shift is larger for bidder i when the probability distribution of the second-lowest cost, conditional on i winning, changes more with entry. This varies with both the initial number of bidders and the cost distribution. Holding the winner's cost constant at the expected minimum cost draw, the distance between the curves isolates portion of the price decrease due to the strategic response.

The second channel is the extensive margin. A larger pool of bidders increases the chance that one bidder draws a low cost. I draw new bidders from the same cost distribution as incumbents, consistent with a model in which potential entrants have similar production capabilities and make the entry decision prior to learning their project-specific cost draw, as in Levin and Smith (1994). This counterfactual therefore isolates the effect of expanding participation among otherwise comparable firms. An alternative model with selection on realized production costs could instead imply that marginal entrants draw from a right-shifted cost distribution, diminishing the gains from an additional entrant.

Strikingly, most of the price reduction does not come from drawing a more productive winner, but from the strategic response of incumbent bidders. This is shown in Figure 6: Subfigure (a) isolates the composition effect, while Subfigure (b) adds the strategic response to show the total effect. The full effect, ΔPrice, combines this strategic effect with the composition effect. This decomposition also highlights a related implication: because $\beta(c)$ flattens in the left tail, much of the gain from an unusually low-cost draw is captured by the firm rather than passed through to the government.



(a) Pool Effect



(b) Total Effect

Figure 6: Decomposition of Effect of Increased Competition

Note: Figure shows estimated bidding strategies under the actual and counterfactual bidder distributions for Alabama (2023, cluster 1). Points are observed bids plotted against estimated costs. The gap between the curves reflects the strategic adjustment holding cost fixed. The total price effect combines this with the shift left in expect cost of a new lowest draw.

I estimate the effects illustrated in Figure 6 numerically for all states using the 2023 cost distribution and baseline bidder counts. Table 7 reports results for the strategic effect alone and the total effect for winning bids. On average, a guaranteed additional bidder yields just over $350,000 in savings per auction. Over a baseline average winning bid of $3.7 million in 2023, the savings represent a 10% fall in prices. Remarkably, this estimate closely matches the quasi-experimental result, despite the

two exercises relying on entirely distinct identification strategies and sources of variation.Comparing the last two rows of the table shows that most of the counterfactual price effect comes from the strategic response, which alone generates nearly $300,000 in savings. Consistent with the flattening of $\beta(c)$ in the left tail, markups fall more under the strategic effect alone than under the total effect: when the additional bidder wins with an unusually low cost draw, it can retain part of that cost advantage as a higher markup.

	All winning bids				
	Auction-level			State-level (means)	
Variable	Mean	25th pctl	75th pctl	25th pctl	75th pctl
Baseline					
Markup	0.37	0.12	0.44	0.17	0.51
Counterfactual					
Strategic effect					
Markup	0.19	0.06	0.26	0.07	0.34
Δ winning bid ($1000s)	-286	-348	-55	-392	-184
Total effect					
Markup	0.33	0.08	0.39	0.18	0.47
Δ winning bid ($1000s)	-358	-427	-62	-494	-265

Note: Table reports the mean and interquartile range of all winning bids for roadwork projects in 2023 under baseline and counterfactual distributions of the number of bidders. Cost figures are expressed in thousands of 2023 USD. The strategic effect adjusts only incumbent bidders' strategies, while the total effect incorporates the probability that the additional bidder may win. When computing means, the lowest and highest 1% of observations by cost are trimmed..

Table 7: Summary of Counterfactual Competition and Prices

The complexity of the interaction between the cost distribution and bidder counts is shown in the changing markup. The strategy effect alone produces a lower markup than the total effect, although both are lower than the baseline. This is due to the that firms that draw a particularly low cost know they have good odds of winning even with a higher markup.

7 Entry Barriers

The preceding sections show that large variable profits persist even though entry is scarce. In this section, I turn from the consequences of low bidder participation to its causes. Large variable profits need not imply large economic profits if firms face substantial fixed costs of participation. In road construction, these costs may reflect general construction requirements, the cost of entering a procurement market, or the cost of preparing and submitting a bid for a particular auction.

I first examine which of these costs appear to matter in practice. I consider whether firms appear to be deterred by costs general to construction, costs incurred separately for each auction, or costs that arise when entering a new *state* market. For the latter, the geographic pattern is especially informative. If nearby demand attracts firms within a state but not across the border, then the relevant barrier is consistent with state-specific administrative policies and institutions rather than production fundamentals.

I then complete the structural framework by nesting the bidding model above within an auction-entry model. Because the bidding model recovers firms' expected variable profits, observed bidder counts reveal the costs firms must overcome to participate. This allows me to estimate auction-entry costs and infer the magnitude of state-market entry costs.

7.1 Evidence on the Sources of Fixed Costs

Some fixed costs in highway construction are inherent to production. Construction machinery, for example, when not rented, is a substantial capital investment.[53] Fixed costs may also arise from the procurement process, in particular, learning or complying with requirements, such as audit and legal expenses required for prequalification. Additionally, firms face potentially large fixed costs at the auction level, including the administrative cost of preparing bid proposals.[54]

I start by examining the role of fixed costs inherent to the construction industry. If these costs are the main factor deterring procurement participation, we would expect firms to begin bidding relatively early in their lives, when fixed capital investment and procurement entry decisions are made jointly. I examine firm age for the subset of bidders matched to Dun and Bradstreet data on Heavy Construction and Building Construction firms with reliable start-year information. I find that infrastructure market participants are typically established construction firms well before their first government bid. The median firm is 17 years old when it first bids in a procurement auction; Appendix Figure A.12 shows the full distribution. Despite the capital intensity of the construction industry, the fixed investments associated with entering construction therefore appear to be sunk

[53] It's difficult to estimate the share of machinery that is rented versus owned, but machinery rental is the third largest category of intermediate inputs in "Transportation structures and highways and streets" in the BEA Input-Output Use table.

[54] Liscow et al. (2023) report that a one-standard-deviation increase in bid-document length is correlated with 16% higher prices.

well before firms decide whether to enter public procurement.

I now turn to fixed costs at the auction and market levels.[55] As detailed in Bresnahan and Reiss (1991), fixed costs generate economies of scale. Greater scale can therefore support more firms even as competition reduces each firm's variable return per unit of production. I use this relationship to test for fixed costs at both the auction and market levels. I measure auction scale using estimated project cost and local market scale using the number of nearby auctions. I then examine whether greater scale increases the number of bidders per auction.

Table 8 reports results. Column (1) shows that doubling project cost increases bidders per auction by 0.05.Column (2) shows that doubling the number of auctions within 100 miles increases bidders per auction by 0.22—about 7% relative to the median of three bidders. Columns (3)–(4) push this logic further by looking across state lines. Remarkably, doubling the number of nearby auctions across state borders has no effect on bidders per auction. This implies that market fixed costs scale with state boundaries. Crossing into another state requires incurring a new entry cost. In short, the fixed costs at infrastructure market level are triggered anew with each new state, suggesting they are the result of regulatory entry barriers.

[55]These costs correspond to "entry barriers" under the classic definition of Stigler (1968): a barrier to entry is a cost of producing that must be borne by firms seeking to enter an industry but not by firms already in the industry.

	Dependent variable:			
	Number of bidders			
	(Project scale)	(Market scale)		
	(1)	(2)	(3)	(4)
Log $\widehat{\text{cost}}$	0.047***	0.051***	0.050***	0.050***
	(0.007)	(0.007)	(0.007)	(0.007)
Log auctions < 100 mi		0.224***		
		(0.017)		
Log in-state auctions < 100 mi			0.239***	0.214***
			(0.015)	(0.017)
Log out-of-state auctions < 100 mi			−0.010*	−0.004
			(0.006)	(0.006)
Log mi to border				0.005***
				(0.002)
Rurality	−0.415***	−0.343***	−0.346***	−0.353***
	(0.018)	(0.019)	(0.018)	(0.018)
Ruggedness	−0.004***	−0.004***	−0.004***	−0.004***
	(0.0004)	(0.0004)	(0.0004)	(0.0004)
Tons (1000s)	−0.002***	−0.002***	−0.002***	−0.002***
	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Type FE	yes	yes	yes	yes
Class FE	yes	yes	yes	yes
year FE	yes	yes	yes	yes
state FE	yes	yes	yes	yes
Observations	36,657	36,657	36,657	36,657
Adjusted R^2	0.224	0.227	0.230	0.230
Note:				*p<0.1; **p<0.05; ***p<0.01

Note: Table reports estimates from regressions of the number of bidders per auction on measures of market scale. The first column looks at project scale, measured by project cost, and the last three columns look at market scale, measured by the number of auctions within 100 miles of a given auction. The 100 mile threshold is chosen based on the entry-by-distance results in Appendix Figure A.9.

Table 8: Market Scale and Fixed Costs

7.2 Quantifying Auction and State-Market Entry Costs

The preceding evidence shows that fixed costs deter entry and, importantly, that market-level costs reset at state borders. The remaining question is quantitative: are the primary barriers incurred each time a firm bids, or when it enters a state procurement market? I answer this by nesting the bidding model of 4 in an auction-entry model. The key idea is simple: the bidding model recovers firms' expected variable profits, and firms enter only when those profits exceed their entry costs. Observed participation therefore reveals the distribution of auction-entry costs. This framework does not attempt to directly measure regulation—a formidable task (Trebbi and Zhang, 2022, Bartik

et al., 2025)—but instead quantifies auction-level entry costs. I then combine these estimates with a zero-profit condition at the state-market entry margin to infer market-entry costs. This allows me to compare the two directly and ask where the larger barrier to competition lies.

I begin by embedding the estimated actual and counterfactual margins from Section 4 in a threshold-crossing entry framework, in the spirit of Berry (1992). Firms draw heterogeneous auction entry costs that depend on firm covariates, a firm-auction-specific shock, and an auction-level common shock. After observing its entry cost, each firm decides whether to enter, taking into account this cost and the expected margins from bidding in that auction. I assume that firms learn their variable production costs only after the entry decision; bid-preparation costs are therefore incurred before firms know their production-cost draws. In the auction literature, this timing assumption follows Levin and Smith (1994), in contrast to Samuelson (1985), where project costs are known prior to entry. The assumption rules out selection into auctions on realized production costs and, in particular, shapes how competitive an additional counterfactual bidder is.

This setting offers two advantages over standard threshold-crossing entry models. First, I am not faced with jointly estimating entry costs and variable profits from market observables: variable profits are disciplined by the bidding model above. Observed participation can therefore be used to recover entry costs without imposing a reduced-form profit specification, such as the linear-in-observables specification common in entry applications following Bresnahan and Reiss (1991). Second, because variable profits are measured in dollars, they fix the scale of the model, allowing entry costs to be identified in dollar levels rather than only up to scale.

I assume the following sequence of events. First, firms in the market draw an auction entry cost. Second, firms make an auction entry decision. I assume the firm with the lowest entry costs enters first, ensuring a unique equilibrium, as in Berry (1992). Finally, firms learn their production cost, and then place their bid. At step one, the expected total auction profit is given by

$$\mathbb{E}\Pi_{ia} = \overbrace{V_a}^{\text{variable profit}} - \overbrace{\phi_{ia}}^{\text{entry cost}},$$

where variable profit is the expected auction margin $V_a = \mathbb{E}\left[(b_{ia} - c_{ia})\,\mathbb{1}\{\text{win}\}\right]$. As before, b_{ia} is firm i's bid and c_{ia} is firm i's production cost, and $\mathbb{1}\{\text{win}\}$ again also depends in particular on $p_N(n)$, the probability mass function of the number of bidders in the auction. The firm thus conditions on the distribution of its number of rival bidders, rather than on the exact value, as before. At the auction entry stage, the expectation is over the firm's own cost.

I parameterize auction entry costs as

$$\phi_{ia} \equiv \kappa_c + \mathbf{Z}_{ia}'\boldsymbol{\alpha} + \beta s_a + \rho_c \mu_{ao} + \sigma \mu_{ia},$$

where κ_c is an intercept that varies by project-type cluster c. The vector $\mathbf{Z}_{ia}$ contains standardized firm covariates, so κ_c can be interpreted as the entry cost for a firm with mean covariate values and no shocks. Specifically, experience is standardized $\log(1 + \text{bids in the preceding five years})$, and geographic scope is the standardized number of states in which the firm participated before 2023. The auction-level covariate s_a is standardized log project scale, where project scale is the component of bids explained by observed project cost shifters, as estimated in Section 4.3. The shocks μ_{ao} and μ_{ia} are independent standard normal variables, common to all potential entrants in auction a and specific to firm-auction pair ia, respectively. Thus, ρ_c and σ are the corresponding shock standard deviations. All dollar-valued parameters and shocks are measured in thousands of 2023 dollars.

I estimate the model on Illinois auctions in 2023. I construct the potential-entrant pool from firms with pre-2023 Illinois bidding histories. For each firm, I use its modal pre-2023 Illinois bidding county as its primary Illinois location and calculate the 90th percentile of the distances it previously traveled within Illinois. A nonbidding firm is included as a potential entrant for an auction if the project lies within that firm-specific radius, the firm has previously won the same project type, and its six-month workload does not exceed the 95th percentile of its own historical six-month workload. All observed bidders are retained regardless of these screens. In the two auctions for which the resulting pool otherwise contains only the observed bidders, I add the nearest otherwise-excluded Illinois firm so that the likelihood comparison with $N + 1$ potential entrants is defined. Appendix Table A.18 reports sensitivity to using smaller and larger potential-entrant pools.

The likelihood of observing N_a firms in auction a can be written

$$\Pr\left(N_a = N\right) = \underbrace{\Pr\left(\left|\{i : \mathbb{E}\Pi_{ia}\left(p_N(n)\right) \geq 0\}\right| \geq N\right)}_{\text{at least } N \text{ profitable under } p_N(n)} - \underbrace{\Pr\left(\left|\{i : \mathbb{E}\Pi_{ia}\left(p_N(n+1)\right) \geq 0\}\right| \geq N+1\right)}_{\text{at least } N+1 \text{ profitable under } p_N(n+1)}. \tag{9}$$

The first equation counts the number of firms with positive expected profits in an auction with the covariates of a and $p_N(n)$, the status quo equilibrium distribution of the number of bidders. The second equation counts the analogous number of firms with expected positive profits under the more competitive counterfactual probability distribution $p_N(n+1)$. In words, the likelihood is given by the probability that at least N firms could be profitable in the observed equilibrium minus the probability that $N+1$ would be profitable if firms shifted to an equilibrium with one additional bidder per auction.

The parameters of interest are $\theta = (\kappa_1, \kappa_2, \boldsymbol{\alpha}, \beta, \rho_1, \rho_2, \sigma)$. Estimation requires a distribution assumption for the cost shocks; I assume that μ_{ao} and μ_{ia} are standard normal and mutually independent. I construct V_a using the estimated bidding model. I draw production costs from the recovered cost distribution, restricted to nonnegative production costs, evaluate the estimated equilibrium bidding function, and calculate expected variable profit by averaging the markup associated with each cost draw, $b(c) - c$, multiplied by the model-implied probability of winning. I construct counterfactual expected profit analogously, using the counterfactual bidding function and shifting the bidder-count distribution to represent one additional bidder. Because these calculations are initially performed using homogenized bids, I rescale the resulting expected profits using each auction's project-specific scale factor.

I estimate the posterior distribution of θ using the independence Metropolis–Hastings procedure described in Appendix C.4. The likelihood integrates over both the auction-level common shock and the firm-auction-specific shocks. Appendix Table A.17 reports convergence and simulation-precision diagnostics, and Appendix Figure A.18 reports trace plots for the four chains.

Table 9 reports posterior means and 95% credible intervals. The posterior mean of the average entry cost across the first three entrants is \$112,000, with a 95% credible interval of \$69,000–\$148,000.[56]

[56]For comparison, several papers using different modeling approaches estimate entry costs ranging between 2% of

These estimates imply substantial costs of entering individual procurement auctions.

Estimated mean costs increase across the ordered entrants: the posterior means are -$111,000, $141,000, and $304,000 for the first, second, and third entrants, respectively. The negative estimate for the first entrant reflects the high dispersion in entry costs needed to explain limited entry from a large pool of potential entrants. Appendix Table A.18 shows that the potential-entrant pool remains sizable under a more restrictive definition.

The estimated costs should be interpreted broadly. The revealed choice bundle at the auction-entry stage includes not only bid preparation, but also engineer hours, legal review of contract terms, bid bonds and pre-qualification work, DBE documentation, and site visits. The estimates may also partly capture capacity constraints, for example when firms near capacity must rely on outside estimators or counsel or require engineer overtime.

	Estimate	95% CI
Implied entry costs		
Mean entry cost, first three entrants	111.5	[68.8, 148.0]
First entrant cost	-110.5	[-196.9, -38.2]
Second entrant cost	141.5	[104.2, 174.9]
Third entrant cost	303.7	[265.4, 347.6]
Structural parameters		
α_{exp} (experience)	-262.2	[-435.1, -74.0]
α_{scope} (geographic scope)	-91.3	[-263.0, 103.6]
β (project scale)	231.3	[188.6, 275.9]
κ_c (cost intercept)		
$c = 1$	993.0	[843.3, 1151.4]
$c = 2$	781.3	[642.6, 926.2]
ρ_c (project-shock SD)		
$c = 1$	123.9	[46.7, 196.6]
$c = 2$	135.6	[86.7, 193.8]
σ (firm-project-shock SD)	472.7	[351.4, 614.3]

Note: Table reports posterior means and 95% credible intervals from the entry model for Illinois in 2023. The first rows report implied entry costs; the first row averages across the first three entrants. Structural parameters include bidder experience (α_{exp}), geographic scope (α_{scope}), project scale (β), the entry-cost intercept (κ_c), project-level shock dispersion (ρ_c), and firm-project shock dispersion (σ). The index c denotes the two auction clusters used in estimation: $c = 1$ contains auctions with relatively few predicted bidders and $c = 2$ contains auctions with relatively many predicted bidders.

Table 9: Posterior Estimates

the engineer's estimate (Bhattacharya et al., 2014), 2–4% of the engineer's estimate (Krasnokutskaya and Seim, 2011), 5–6% of the engineer's estimate (De Silva and Rosa, 2024), and 8% of the bid (Li and Zheng, 2009).

Despite this broad interpretation of auction entry costs, the estimates indicate that market-level entry costs, to my knowledge not previously quantified in this literature, are the dominant barrier to competition. I compare auction entry costs to market entry costs by imposing a zero-profit condition at the state–market level. This assumption is plausible in a setting with many active firms, and as shown in Table 1, the average state–year includes 84 unique winning firms. With a large number of firms, an entry model with heterogeneous firm-level entry costs yields tight bounds on the wedge between market fixed costs and variable profits.[57] This observation suggests a straight-forward back-of-the-envelope calculation. I estimate annual total auction-entry costs by multiplying the average per-auction entry cost for the first three bidders, $112,000, by the average number of auctions a firm enters per year, 16.7. This gives annual auction-entry costs of approximately $1.9 million per firm. I compare this amount to annual market-entry costs of approximately $9.2 million per firm, estimated from the zero-profit condition. Auction-entry costs therefore account for about 17% of total fixed costs, leaving about 83% attributable to market-entry costs. Despite prior attention to auction-entry frictions, the evidence here points to substantially larger market-entry costs, implying that fees, procedural requirements, and the learning required to work with the government are the primary barriers to entry.

8 Conclusion

U.S. transportation demand is growing while the physical system is aging, making it increasingly important to understand how to reduce the costs of reconstruction and upgrades. The lessons learned may also be relevant for other types of public infrastructure, such as the electricity grid, which is lagging far behind goals. This paper focuses on one long-hypothesized but understudied channel: competition in the market for infrastructure construction. In particular, I examine the role auction competition has played in raising prices above firms' costs of production, focusing on the direct impact of the number of auction participants on bids. Yet market structure may further affect prices in the long run by shaping incentives for innovation — my results may be an underestimate of the

[57] The market entry condition is $V(y_m + 1) < F_m \leq V(y_m)$, where y_m denotes the observed number of active firms in market m, $V(y_m)$ is the variable profit per firm at that number of firms, and F_m is the fixed cost of market entry. If $\frac{\partial V(y_m)}{\partial y_m} < 0$ and $\frac{\partial^2 v(y_m)}{\partial y_m^2} > 0$, then as y_m increases the bounds tighten, and for sufficiently large y_m we have $v(y_m) \approx F_m$.

long-run consequences of limited competition.

This paper documents that weak competition is a defining feature of U.S. infrastructure procurement. Across the country, auctions routinely attract only a handful of bidders, and estimated markups are large and have been rising since 2014. These markups account for more of the price growth over the last decade than rising production costs do. I show that securing one more bidder per auction would yield substantial returns to the government, a result that I establish with both a quasi-experimental design and a structural model with separate identification strategies. Yet, entry remains scarce. Entry patterns are consistent with the existence of sizable fixed costs generated by regulatory burdens rather than physical investments. These entry costs deter participation in both individual auctions and in the market itself; they appear largest at the state-market level. While reducing regulation and bureaucracy would reduce prices, in some cases doing so may be difficult or undesirable, as regulations have benefits. In such a case, increasing the scale of either auctions or market size would grow variable profits relative to fixed entry costs and reduce markups. These results motivate further research on barriers to entry and on other policy approaches, from informational outreach to interstate coordination, to improve procurement outcomes.

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A Supplemental Figures and Tables

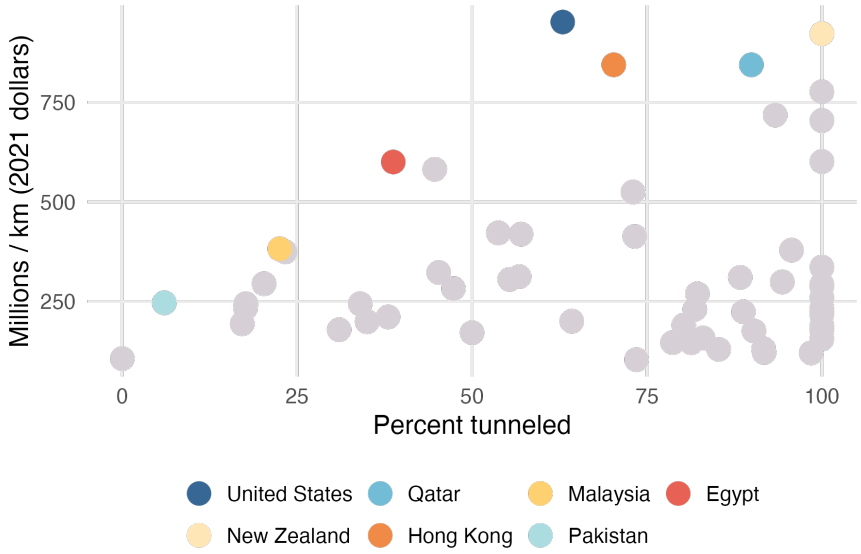


Figure A.1: Mean spending per kilometer over complexity (mean percent of project tunneled)
Note: Figure plots the mean spending per kilometer over mean percent of project tunneled for 59 countries. Country averages are taken over projects, so that each project is weighted equally. When weighting each project by length, the U.S. falls to 9th most costly. Data is from the Marron Institute at NYU.

State	Mean	Median	Mean %	Median %	Source
Massachusetts	-26, 990	15, 653	8.46%	1.67%	Bolotnyy and Vasserman, 1998-2015
California	457, 746	58, 168	4.32%	3.25%	CALTRANS, 2009–2021
Colorado	169, 165	37, 482	3.7%	2.1%	CDOT, 1999–2023
New York City	-13, 052	-77, 648	-2.14%	-3.87%	NYCDOT, 2009–2021

Note: Table shows the cost overrun (final price minus winning bid) for four states. Data for Massachusetts comes directly from Table 1 in Bolotnyy and Vasserman, 2023 and covers only bridges. The other data comes from the relevant DOTs. The sample sizes are 1651, 2256, and 1265, for California, Colorado, and New York, respectively. For the latter three states, estimates control for inflation and are in 2023 dollars .
Table A.1: Cost Overruns for Four States

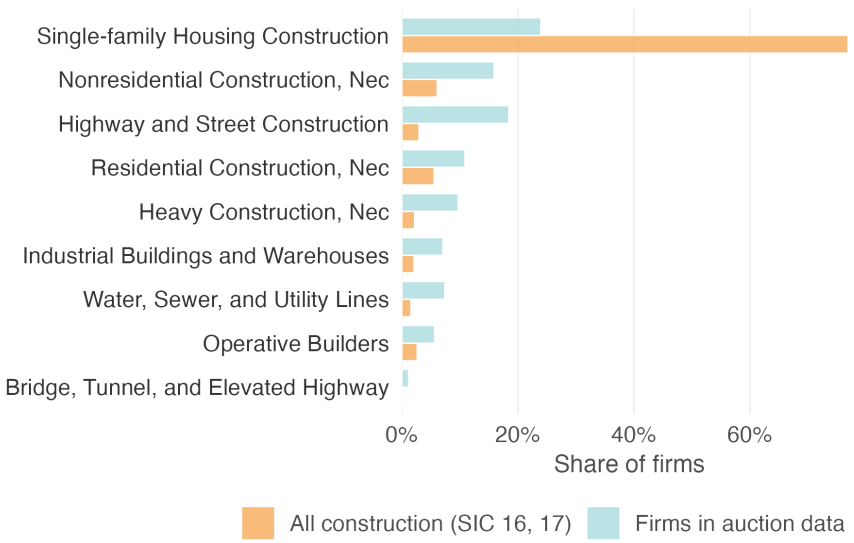
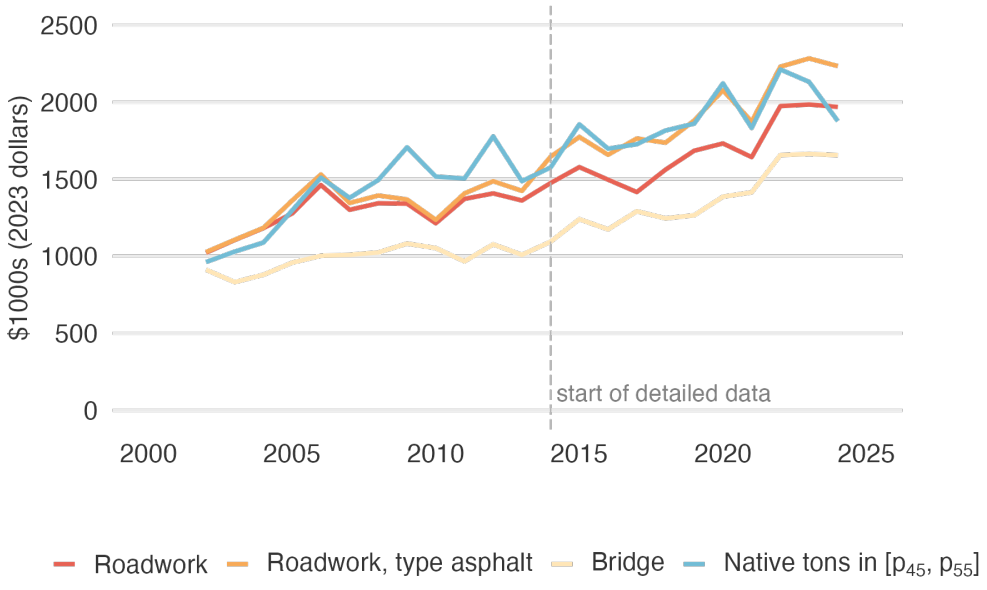
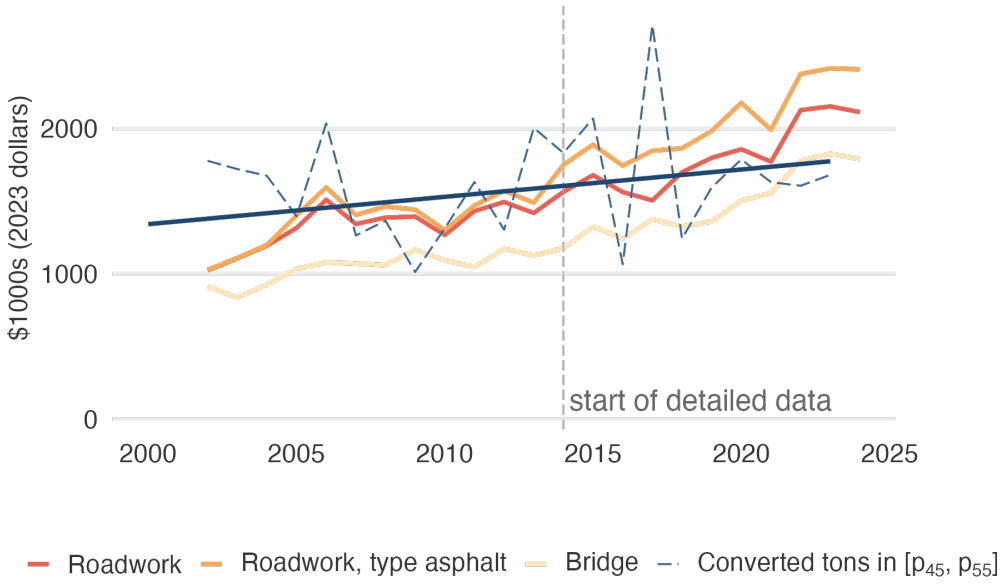


Figure A.2: Shares of Infrastructure Firms and All Firms by 4-Digit SIC Code (SIC 15–16)
Note: Figure plots the distribution of firms across four-digit SIC codes. "Firms in auction data" are those matched to the DOT auction sample by name and state, while "All firms" are Dun & Bradstreet firms in the broader SIC 15–16 construction categories. Each unique firm name is counted once. A small number of firms have multiple SIC codes; shares therefore computed by weighting each firm's contribution equally across its distinct SIC codes.



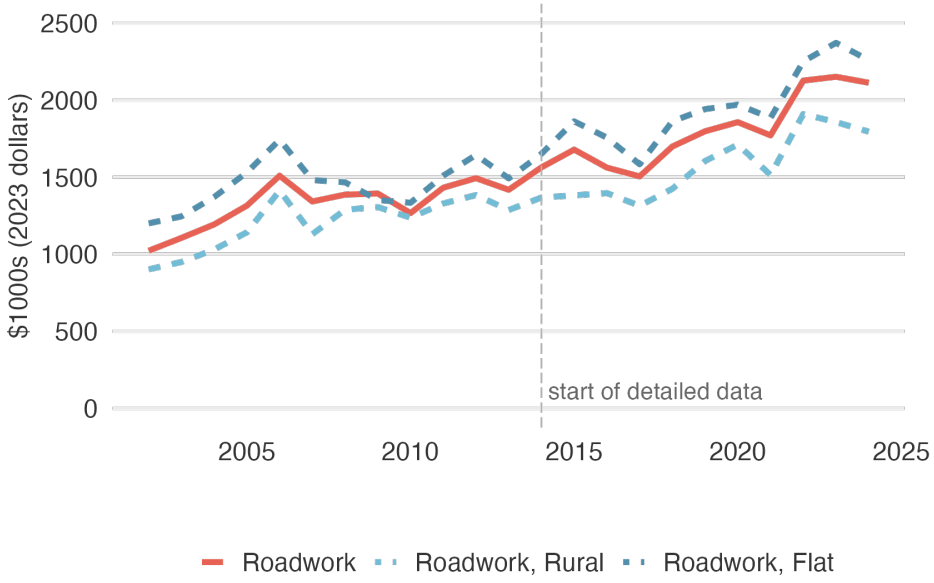
(a) Balanced Panel



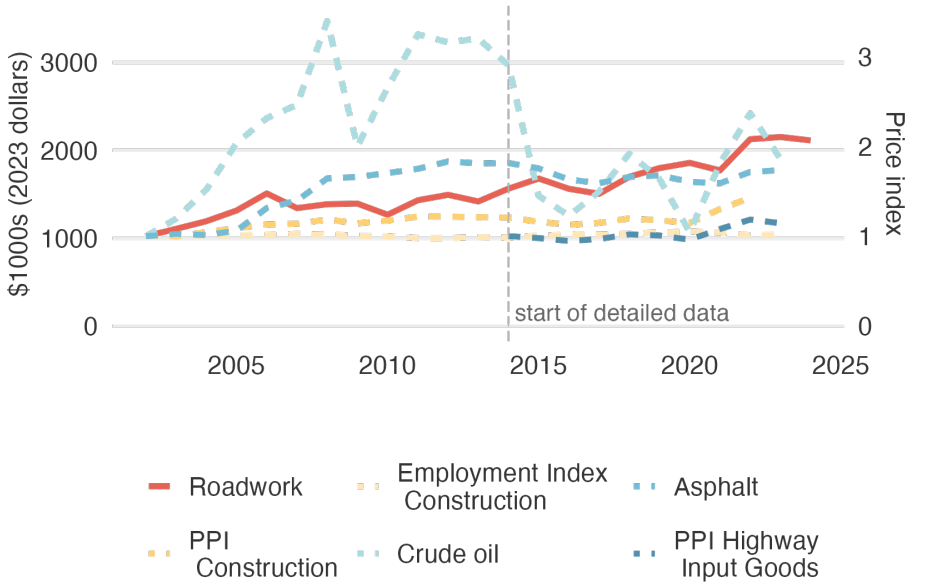
(b) Trend with Converted Tons

Figure A.3: Price of Infrastructure Over Time Robustness

Note: Panels plot the median winning bid of DOT infrastructure projects over time, as in Figure 1. All prices control for inflation and are in 2023 dollars. Panel (a) replicates Figure 1 but uses a balanced panel of states, which includes 41 states. In Panel (b), the dark dashed blue shows the same exercise for the alternative measure of size, converted tons, for the six states that have bid-schedule data since 2000. Construction of the converted tons measure is described in Appendix Section B.4. The sold dark blue line shows the jackknife bias-corrected linear trend estimated on data from 2000 to 2024 for the dark-blue series, based on leave-one-state-out resampling. Leave-one-out slopes have large variation: $200, $18,000 $19,000 $21,000 $23,000 and $ 64,000 per year, and with only six states the 95% jackknife confidence interval includes zero.



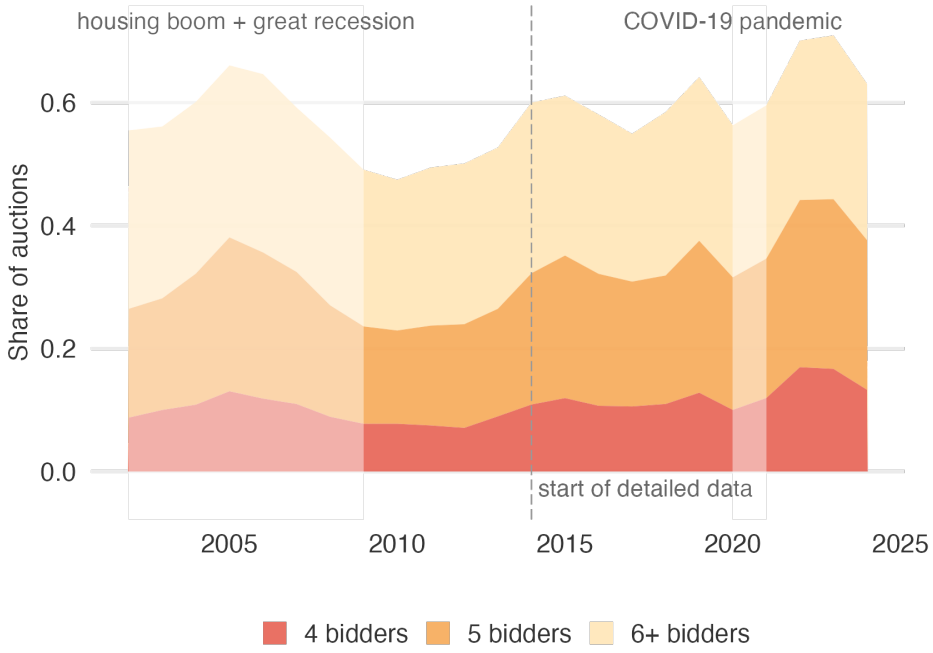
(a) Price and Location



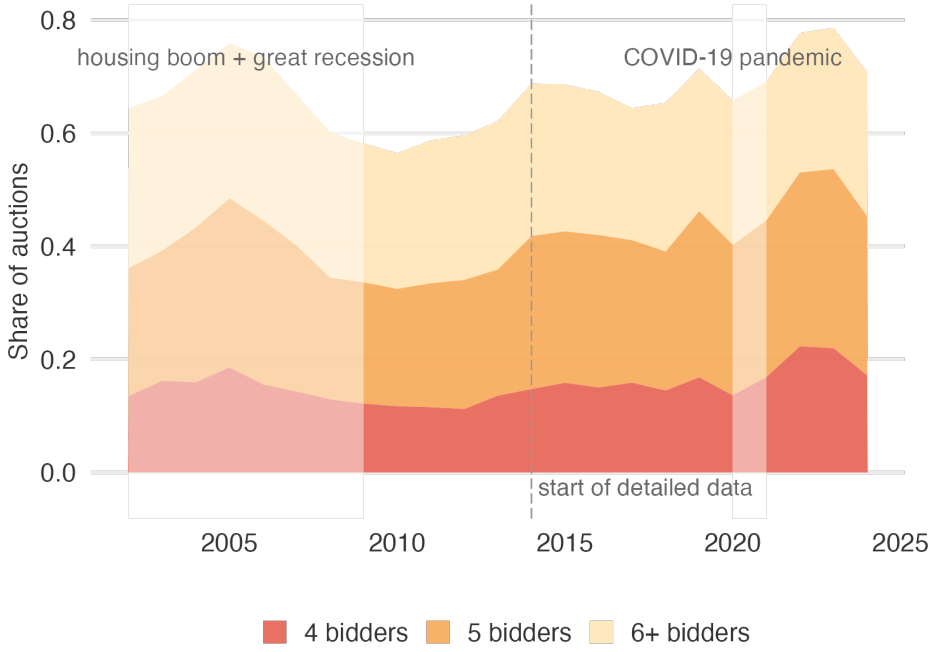
(b) Price and Input Prices

Figure A.4: Price of Infrastructure Over Time

Note: Figures plot the median winning bid for roadwork auctions over time, as in Figure 1. Figure (a) overlays the roadwork median with median prices for projects located only in rural counties and only in flat-terrain counties. Figure (b) overlays the roadwork median with three input-cost indices: (i) the Construction Producer Price Index (PPI), (ii) the Employment Cost Index (ECI) for construction labor, (iii) the national average crude oil price, the main feedstock for asphalt, and (iv) the PPI for Net Inputs to Highways and Streets, Goods. This index is only available since 2015. All four indices are also divided by the CPI for comparability.



(a) Number of Bidders Over Time, Balanced State Sample



(b) Number of Bidders of Time, Roadwork Auctions Only

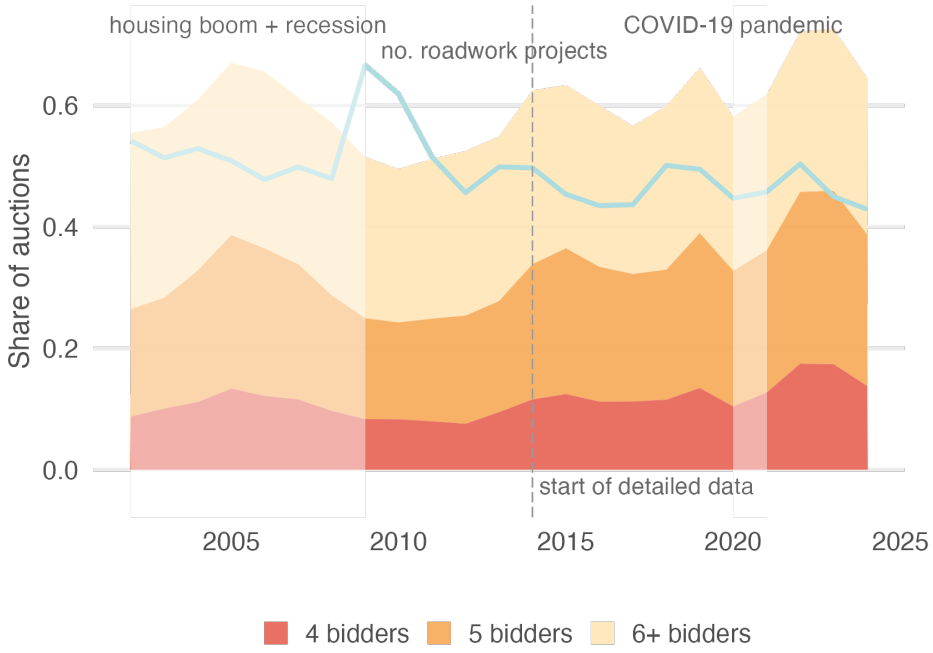
Figure A.5: Macro-economic Effects and Decreasing Bidder Trend: Robustness

Note: Figures plot the share of auctions by number of bidders over time, as in Figure 2. The dashed vertical line marks the beginning of the detailed dataset used later in the paper, indicating the period covered by the structural estimation.

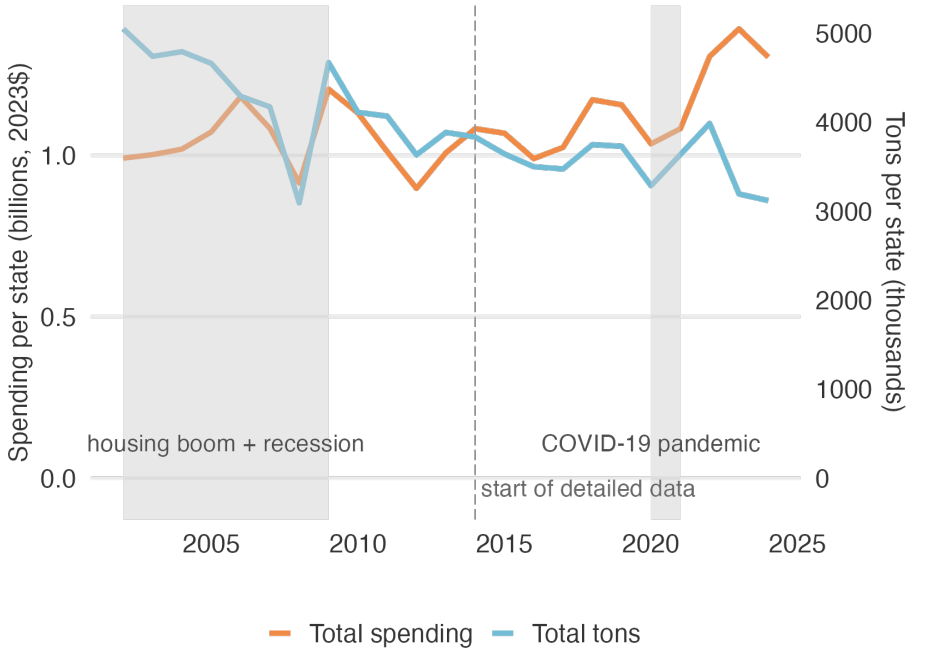
Metric	State Market Entrant	Procurement Market Entrant
Share of bids	3.3%	2.4%
Share of wins	3.0%	2.4%
Share of value	4.9%	3.8%

Note: Table reports the share of bids, wins, and total contract value captured by firms in the year they enter the market. Data begin in 2010 to include all states while allowing for at a minimum three-year lead-in period, to avoid misclassifying firms' first appearance in the data as true market entry.

Table A.2: New Entrants Account for Only A Minor Share of Market



(a) No. of Bidders Over Time with Total No. of Projects



(b) Total Spending and Total Size (Tons) Over Time

Figure A.6: Flat Infrastructure Demand Over Time, Though Rising Spending

Note: Figures (a) plot the share of auctions by number of bidders over time, as in Figure 2, along with the total number of projects each year. Figure (b) plots the total spending each year, along with the total size of projects, as measured by total tons of inputs.

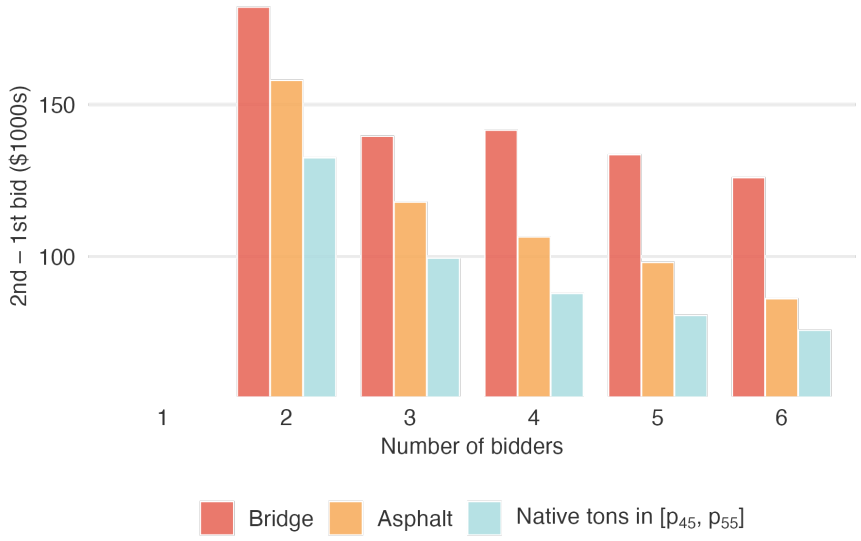


Figure A.7: 2nd - 1st Bid Gap

Note: Figure plots the national median difference between the second lowest and lowest bids by number of bidders in the auction for all bridge, asphalt roadwork, and roadwork projects within a narrow range of native tons, respectively. To account for state omitted variables that effect bid gaps and number of bidders, gaps are demeaned by state with the national mean added back. When taking means, 1% of the sample is trimmed due to large outliers.

	Dependent variable:			
	Winning Bid			
	Road projects	Asphalt only projects	+ Size	+ Complexity
California	9,222.78***	6,471.71***	4,547.30***	1,733.85***
	(536.85)	(520.28)	(493.51)	(514.73)
Log Tons of Asphalt			2,961.60***	2,730.33***
			(144.37)	(233.22)
Asphalt Share of Price				−9,231.65***
				(1,975.57)
Log No. items				2,169.28***
				(614.53)
Constant	1,463.40***	1,530.78***	−23,295.48***	−21,997.62***
	(250.08)	(247.51)	(1,231.98)	(1,879.30)
Observations	3,613	2,744	2,744	2,744
R^2	0.08	0.05	0.18	0.24
Note:				*p<0.1; **p<0.05; ***p<0.01

Note: Table shows the difference in average winning bids between California and Kentucky for projects from 2018 to 2024. The outcome is normalized by CPI and is in 2023 dollars.

Table A.3: Difference in Prices: California vs. Kentucky

	Pooled correlation	
Covariate	Estimated coefficient	(SE)
Log avg. petrol product price	0.991***	(0.286)
Log construction wage	-0.006	(0.044)
Pct unionized	-0.010***	(0.002)
State prevailing wage law	0.107**	(0.045)
Log population	0.001	(0.022)
Log median HH income	0.041*	(0.024)
Log construction establishments	1.138***	(0.077)
Log firms in market	-0.365***	(0.029)
Herfindahl–Hirschman Index (0–1)	1.445***	(0.229)
State FE	no	
Year FE	yes	
Observations (by row): 927, 927, 927, 815, 927, 927, 927, 927, 927		
Adj. R^2 (by row): 0.075, 0.063, 0.086, 0.050, 0.063, 0.066, 0.246, 0.206, 0.102		

Note: Table reports estimates from a log-linear regression of prices (log winning bids) on covariates, analogous to Table 3 but without state fixed effects. Given project design and engineering vary substantially by state, the correlations are difficult to interpret.

Table A.4: Price Growth and Covariates

	Dependent variable:		
	Δ Log Bid (County First-Differences)	Log Bid (IV: 75 miles)	Log Bid (IV: 125 miles)
No. bidders	−0.014***	−0.093	−0.215***
	(0.003)	(0.065)	(0.068)
Log tons	0.133***	0.136***	0.133***
	(0.002)	(0.002)	(0.002)
State FE		yes	yes
Time Trend	yes	yes	yes
Lag log bid	yes		
First-stage partial F		55.2	59.1
Observations	19,950	22,050	20,728
Adjusted R^2	0.430	0.485	0.407
Note:			*p<0.1; **p<0.05; ***p<0.01

Note: Table reports estimates from modified regressions from those reported in Table 4. Column shows results from a first differences model: $\Delta \log(\text{price})_{c,t} = \gamma \Delta N_{c,t} + \Delta x_{c,t} + \mu_t + \Delta \epsilon_{c,t}$, where $\Delta y_{c,t} \equiv y_{c,t} - y_{c,t-1}$. Columns 2-3 report IV results analogous to column (6) in Table 4, but adjusting the threshold distance between the auction and the firm location.

Table A.5: Estimates of the Impact of Number of Bidders on Prices

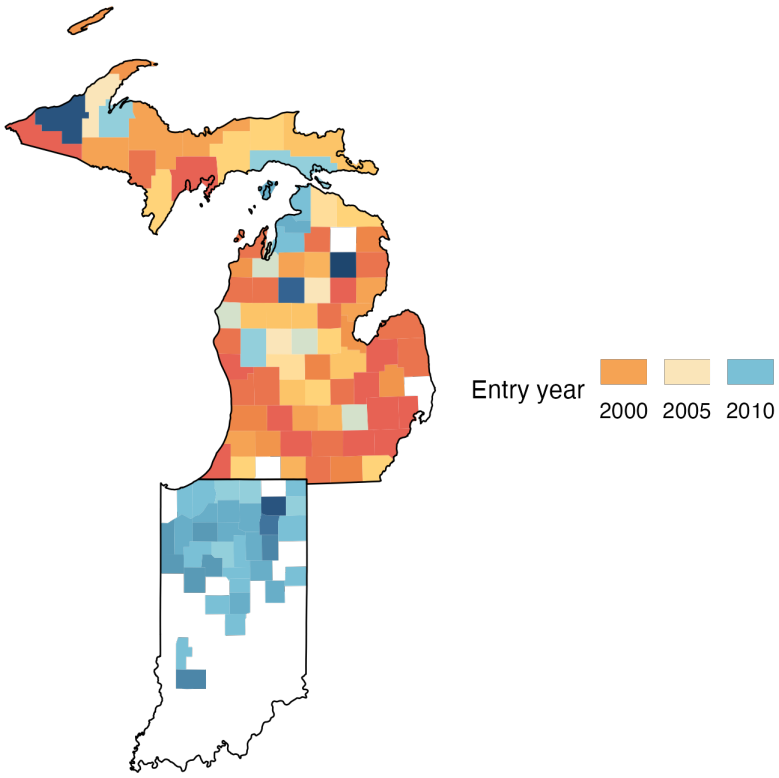


Figure A.8: Example of a firm in expanding to a new state

Note: Figure plots the expansion of one representative multi-state contractor across counties. Shaded areas indicate the first year the firm bid on a project in each county, with lighter colors corresponding to earlier years.

	Means				
trim	Cost	Margin	Cost (homogenized)	Margin (homogenized)	Markup
0%	2603	870	1493	801	1.31
1%	2477	693	1593	669	0.22
3%	2243	604	1638	576	0.22
5%	2079	554	1643	528	0.23

Note: Table reports the means of model estimates of costs, margins, and markup, by the percent of observations that are trimmed from each tail of the cost distribution. The markup stabilizes after 1% of the sample is trimming.

Table A.6: Means of Model Estimates by Tail Trim Percent

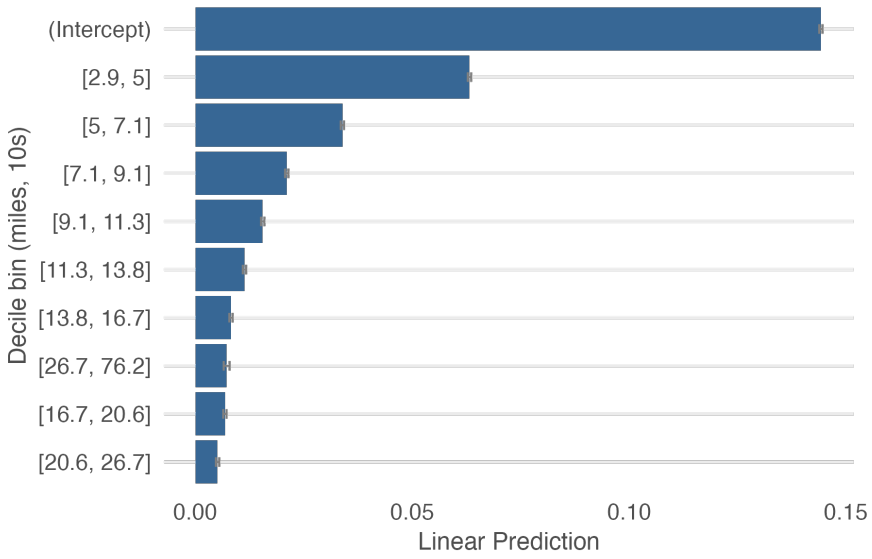


Figure A.9: Estimates from Regression of Auction Entry on Distance Bins

Note: Figure plots predicted probabilities (point estimates plus intercept) from a linear probability model of a firm entering an auction, as a function of the distance between the firm's county and the auction's county. Firm locations are taken from a matched subsample of Dun and Bradstreet's heavy construction firms. The 6% of auctions lacking county identifiers—for example, those designated as "state-wide"—are dropped. An active firm is defined as one that submitted a bid in any auction during the year, or in both the preceding and following year. All auction types are included.

B Data Appendix

B.1 Auction data

For each state year, I have a file with all lettings for the year. For each letting, I extract the job id, the county, the letting date, a description of the job, the total tons of input required for the job, the engineer's estimate when available, the bidder name and id, the bidder rank, the bid, and the project engineering "type," a variable pre-defined by the data provider as the most expensive input category of item in the project.

States vary by their start year in the sample and their data coverage for the project description and the engineer estimate. Data is missing when either the state does either does not record it or does not share it with the data provider. Finally, while the tons variable is always present, states vary in the share of projects with inputs measured in tons; this reflects both variation in project type and in chosen units. Table A.7 below gives the min, max, 25th, 50th, and 75th percentile for each of these variables, as well as the number of auctions per state. Additionally, all states except Rhode Island

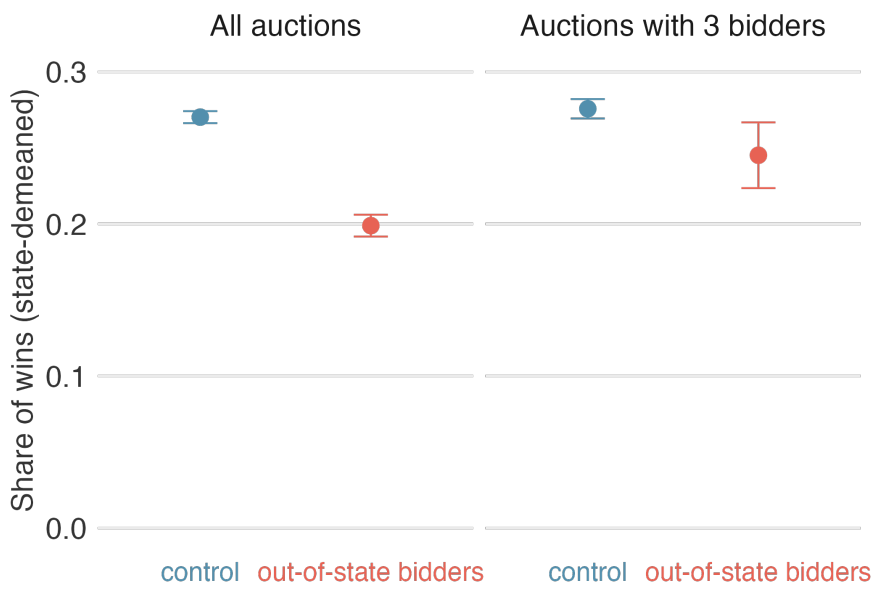


Figure A.10: Estimates from Regression on Indicator for Out-of-State Bidder

Note: Figure plots the mean share of wins for out-of-state entrants and the control group after removing state means; whiskers show 95% confidence intervals. Estimates come from OLS on the state-demeaned outcome, shifted back by the control-group mean. The left panel includes all auctions, while the right panel restricts to auctions with exactly three bidders to control for strategic effects. Home-state observations for entrants are excluded, and the unit of observation is firm by state.

provide county information.

variable	min	p25	p50	p75	max
Start Year	1993.00	1996.75	1998.00	2001.00	2007.00
Auctions per year	37.00	99.00	214.00	346.00	952.00
Description	0.61	0.99	1.00	1.00	1.00
Tons > 0	0.44	0.68	0.72	0.77	0.93
Engineer Estimate	0.00	0.00	0.00	0.68	1.00

Table A.7: Data coverage quantiles

For much of the analysis, I restrict to sample of projects identified as “roadwork". I define these projects as projects of engineering type as asphalt or concrete pavement, or with descriptions containing specified roadwork keywords: pavement, grinding, paving, resurfacing, milling, route, or roadwork. I exclude projects of engineering type bridge. This rule is ultimately very similar to using the “roadway resurfacing and repair" LLM classification based purely on the description, described

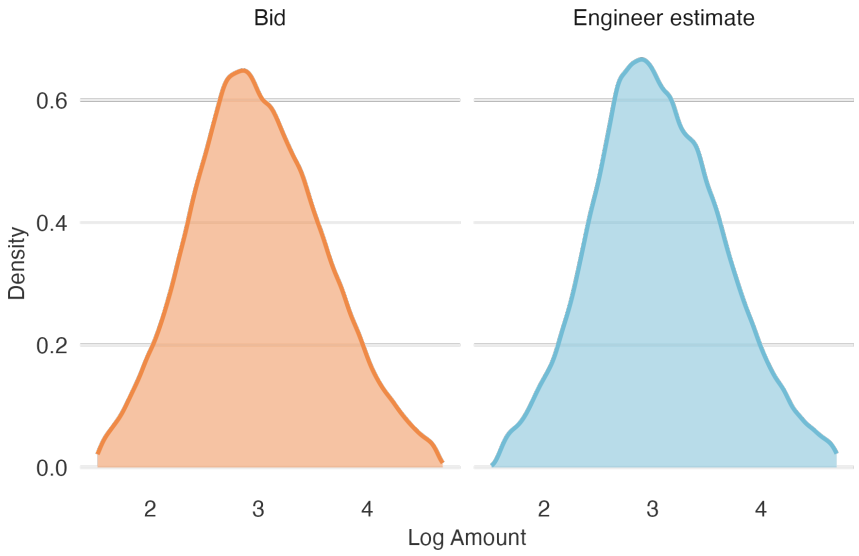


Figure A.11: Bid Density versus Engineer Estimates

Note: Figure plots kernel densities of bids and engineer estimates in log amounts (of thousands of dollars). The left panel shows bids and the right panel shows engineer estimates. The sample covers the 18 states with engineer estimates in the data. The top 1 and bottom 1 percent of bids are dropped.

in Section B.3. Roadwork auctions account for 47% of all auctions.

B.2 Firm data

The auction data include firm names and a within-state identifier, but no additional information on the firms. To illustrate firm industries, Table A.9 reports self-descriptions and listed sub-industries for a random sample of five firms.

Multi-state firms To identify multi-state firms, I use a three-stage process. First, I clean bidder names using standard procedures. Second, I build candidate pairs with fuzzy string matching. Specifically, I apply a constrained Jaro–Winkler (JW) procedure: require overall JW distance 0.20 after cleaning; require at least one shared meaningful word; require the first meaningful word to share the same initial; if that first word has ≤ 4 characters, require an exact match; if it has > 4 characters, allow a first-token JW ≤ 0.08. This modified rule outperforms a simple JW cutoff and still shrinks the candidate set substantially. Third, I classify the remaining non-exact pairs with a large-language model (OpenAI Chat Completions, gpt-5). Each request includes the two firm names, their states, and a fixed prompt requiring a strict "Yes" or "No" on whether the names are the same company, with a confidence score. I construct firm identities by taking the transitive closure

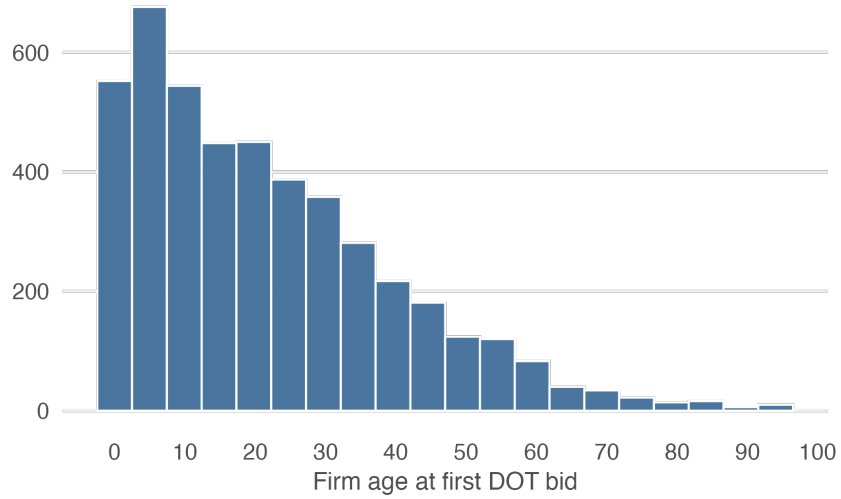


Figure A.12: Firm Age at Infrastructure Market Entry

Note: Figure plots a histogram of firm age at the time of the first infrastructure auction bid, for the 50% of auction participants matched to Dun Bradstreet records in Heavy or Building Construction (see Appendix B.2). To reduce measurement error, the sample is restricted to firms that (i) bid in their matched state of location, (ii) have a first bid at least three years after the auction dataset's start date for that state, and (iii) have a Dun & Bradstreet "start year" prior to the first year the database records the firm. Ages are trimmed to the 0–100 year range, with values outside this interval treated as noise, dropping about 5% of observations. The final sample covers 17% of infrastructure firms. Relative to unmatched firms, they are slightly more active bidders, participating in 28 auctions on average compared to 18, though they win slightly less often, at 23% compared to 27%.

over all positive matches, assigning each equivalence class of names a unique identifier. A firm is classified as multi-state if its identifier appears in more than one state.

Table A.8 reports validation on 100 candidate pairs, split above vs. below the 25th percentile of the model's confidence. I determine ground truth by searching each firm online and checking websites and state records. Reassuringly, I see little evidence that extremely similar cross-state names belong to different firms. Precision is high – 90% in the higher-confidence set and 93% in the lower-confidence set – while accuracy is high in the former and lower in the latter. That's by design: the false-positive rate stays low (3% and 5%). The benefit is that identified interstate links are reliable; the cost is a slight under-count. I quantify the under-counting with a back-of-the-envelope exercise: fuzzy matches contribute 22% of total multi-state firms; the model finds 48% of true matches in the most-uncertain quartile and 75% in the most-certain three-quarters; overall that's 68% of true positives, implying the total multi-state count is under-estimated by about 10%.

Firm covariates To obtain firm locations and start years, I use Dun & Bradstreet, a commercial firm

	High-confidence	Low-confidence
Metric	Top 75%	Bottom 25%
Accuracy	92.0%	68.0%
Precision (PPV)	90.0%	93.3%
False positive rate	2.6%	4.8%
N	50	50

Note: Table reports validation for the LLM's classification of candidate firm pairs as same firm vs. different firms. Pairs are pre-filtered by a Jaro–Winkler (JW) similarity threshold. Panel A randomly samples 50 pairs from the top 75% by the model's confidence; Panel B randomly samples 50 pairs from the bottom 25%.

Table A.8: Validation of interstate firm identities

analytics provider, and pull all firms in SIC 16 ("Heavy Construction, Except Building Construction Contractors") and SIC 15 ("Building Construction—General Contractors and Operative Builders"). I standardize names and first link exact cases. For non-exact cases, I search candidate Dun & Bradstreet firms in the same state or in contiguous states and apply a constrained Jaro–Winkler procedure: overall distance 0.06 after cleaning; require at least one shared meaningful token; require the first meaningful token to share the same initial; if that first token has 4 characters, require an exact match; if it has > 4 characters, allow a first-token Jaro–Winkler 0.07. Stronger restrictions on the first token prevent false matches between companies that are technically similar in spelling but clearly different firms, e.g. "PJ Construction Company" and "KM Construction Company". When multiple candidates satisfy all rules, I retain the candidate with the smallest overall Jaro-Winkler distance. Exact links cover 29% of auction firms in SIC 16 and 43% pooling SIC 15 and 16; the full procedure yields a 48% match rate. A hand audit of 60 randomly sampled pairs finds 95% correct (Table A.10).

B.3 Project Description Based Classification

I classify auctions based on brief project descriptions using OpenAI's GPT-4o-mini language model (March 2025 release) via the chat.completions.create API. The 9 categories given to OpenAI to classify on in Table A.11 below.

Firm	State	Self-description	Listed sub-industries
SCS Engineers	California	SCS is an employee-owned environmental consulting and construction firm that designs and implements sustainable environmental solutions.	Solid and hazardous waste management; Renewable energy; Remediation; Carbon capture.
Dan R. Dalton Inc.	Colorado	Dan R. Dalton Inc. is North America's heat straightening specialists for repairing damaged steel bridges and container cranes.	Railroad bridge repair; Highway bridge repair; Crane repair; Ship repair.
Cutting Edge Group LLC	New York	We specialize in civil construction and design work. We also perform mechanical and electrical work to meet our client needs.	Heavy civil; Building; Parks & recreation; Horizontal directional drilling.
Rutledge Excavating Inc	Pennsylvania	Rutledge Excavating prides itself on delivering top-notch full site development services tailored to meet your needs.	Land clearing and demolition; Concrete work; Erosion control; Stormwater management; Drainage systems; Utility installations; Road construction.
Dome Corporation of America	Washington	We do more than build spaces—we create environments where your team can grow, thrive and innovate.	Capital project feasibility study; Preconstruction; General construction; Commissioning & turnover; Modernization.

Note: Table reports names, state, self published business descriptions, and listed sub-industries for five randomly sample firms in the data. Firms without public websites were excluded.

Table A.9: Firm Profiles

	Validation	
SIC group	Share correct	N
Building construction (SIC 15)	96.7%	30
Heavy construction (SIC 16)	93.3%	30

Note: Table reports share of correct firm matches for 60 randomly sampled firms, stratified by two-digit SIC code.

Table A.10: Validation accuracy of Firm Name Match

B.4 Measuring Project Size

An accurate measure of project size is important for comparing project prices. The data includes the total tons for every project, which is a natural measure of size. However, this measure only includes items that whose unit is tons – projects may differ in which units they use for similar inputs,

Category	Definition
Roadway Surfacing and Repair	Projects involving milling, grinding, resurfacing, and overlaying of roadways, including hot mix asphalt resurfacing, rubberized asphalt overlays, and bituminous resurfacing.
Bridge Construction and Repair	Bridge replacements and rehabilitations, high-rise bridge repairs, painting, deck repairs, and waterproofing.
Culvert and Drainage	Culvert repair and replacement, stormwater pond maintenance, drainage improvements, and minor drainage structure work.
Safety and Traffic Improvements	Traffic signal upgrades, installation of cable barriers, new or replacement guardrails, traffic signs and devices, and traffic control improvements.
Grading and Earthwork	Clearing and grubbing, grading, slope protection, slide repairs, and other work altering the contour of land.
Pedestrian, Sidewalk, and Cycling Infrastructure	Construction or improvement of trails, bikeways, sidewalks, curb ramps, curb and gutter, and safety features for non-motorized users.
Environmental and Landscaping	Environmental mitigation, tree trimming, landscaping, mowing, and other activities to improve or protect the environment.
Facilities	Construction, renovation, or maintenance of buildings, welcome centers, rest areas, park-and-ride structures, weigh stations, or other facilities.
Unknown	Category not specified.

Table A.11: Project categories and definitions.

particularly across states.

I use the detailed data subsample from 2014 to 2023 to show that tons is a good proxy for size *within* state, but less so between states. I then use this data to construct a nationally comparable measure of size for projects between 2024 and 2019.

Table A.12 shows that within states, log tons of a project is highly correlated with other totals of inputs by unit, with an R^2 around .6. However, without state fixed effects, the R^2 drops substantially, in some cases close to 0. This suggests that within a state, the amount of inputs measured in tons and the amount of other inputs are complements. Across states, however, they substitute each other, suggesting states have different unit preferences for the same inputs.

I construct a cross-state measure of project size as total asphalt tons. Using the 2014–2023 item-level subsample, I restrict to projects classified as asphalt. Within those projects, I then flag the items that are truly asphalt—keywords such as asp, hma, hot mix, plant mix, bitum, superpave, or course layers (wear, intermediate, surface, leveling, binder)—while excluding removal, milling, grinding, saw-cut, and coating lines. Each flagged item is converted into short tons using standard engineering factors:

Unit	β	p	Adjusted R^2
Panel A: with state fixed effects			
Log Square Yards	0.2428	0.000	0.623
Log Linear Feet	0.3161	0.000	0.625
Log Each	1.6802	0.000	0.629
Log Lump Sum	1.0968	0.000	0.597
Panel B: without state fixed effects			
Log Square Yards	0.1259	0.000	0.018
Log Linear Feet	0.3848	0.000	0.103
Log Each	0.5115	0.000	0.008
Log Lump Sum	0.6586	0.000	0.012

Note: Table shows coefficients from regressions of the indicated size measure on log tons. Tons are measures as the sum of amounts of all inputs in the project whose unit is tons.

Table A.12: Correlation between Log Tons and Other Measures of Project Size

one ton = 1.0, a metric ton = 1.1, a hundredweight = 0.056, a pound = 0.0005, a gallon of liquid asphalt = 0.00425, a cubic yard of compacted hot-mix = 1.96, a square yard of a 1.5-inch overlay = 0.0825, and a square foot under the same assumption = 0.0092. Summing these conversions at the project level yields total asphalt tons, along with the share of each project's inputs and costs that come from asphalt items. The resulting measure is more stable across states than the raw "tons" field, which reflects only items originally recorded in tons and therefore misses variation in unit conventions.

B.5 Measuring Distance to Firms

I construct a measure of the distance between project locations and bidding firms for the instrumental variable regression with first stage in equation 3. I use the auction data and the procedure described in Section B.2 to identify multi-state firms. I assign each firm's home state h as the first state in which it appears. I define "state entrants" for state s and year y as firms active in a state other than their home state that year: i is an "entrant" if $|\text{bid}_{ist}| > 0, h_i \neq s$. I obtain firms' county locations from the matched sample with Dun & Bradstreet data (see Section 2.2). I use only matched firms in the Heavy Construction industry for this exercise, dropping Building Construction Firms, as the latter is a much larger industry and therefore has a higher probability of duplicate firms names. This match covers 60% of state entrants. I drop firms that "enter" in the first year of data for each state. Finally, I clean the sample by restricting the sample to firms with locations in the identified home

state and where said home state is a neighbor to the target state. Finally, I remove firms that ever establish a location within the target state. The final match covers 13% of entrants.

Each project is assigned a location based on either the centroid of its county or, if multiple counties are involved, the centroid of those counties. I exclude the 6% of projects referenced only by "District," "various," or "statewide." The distance between a firm and a project is defined as the distance from the project location to the firm's nearest establishment. Distance is calculated as the miles between the county centroid of firm location and the county centroid of the auction locatio

C Model details and estimation

C.1 Collusion

Collusion is a natural concern in government procurement. Theory suggests that first-price sealed-bid auctions are the auction format most resistant to collusion, nonetheless the repeated interaction and a potentially limited set of bidders can still create scope for coordination. In most U.S. states bidder identities and bids are disclosed after the auction, which can facilitate coordination across repeated interactions (Hendricks et al., 2003; Liu et al., 2022). Although no definitive test for collusion exists, I present three pieces of evidence indicating that collusion is unlikely to be the primary driver of prices in this market.

Dynamic Competitor Set Collusion in FPSB auctions is sustained dynamically, and so the relevant number of firms is not the number in a given auction, but *over time*. More competitors generally make collusion harder to sustain. I report county-level statistics, as geographic markets may be more granular than at the state level. Counties are likely much smaller than the true geographic markets, so should reflect an underestimate of competition. I restrict to counties with at least ten auctions over a five-year window.

Table A.13 shows results. Counties have a moderate number of active firms. A county meeting the 10 auction threshold has, on average, 31 unique bidders over five years. Limiting attention to important subsets – large auctions and asphalt auctions – reduces the number of bidders to 26 and

12, respectively. However, the latter category simply has few auctions in total.[58] At the firm level, the average firm bids against 32 unique competitors over its lifetime. Excluding firms that bid but never win raises this to 45. The 25th and 75th percentiles are 9 and 57. Overall, these numbers suggest that although firms may face few bidders in the static game, they usually compete against more than a handful of rivals in the dynamic game.

Measure	Mean	25th pct.	75th pct.
Overall			
No. Auctions	27.2	13	30.0
No. Firms	30.6	18	38.0
Big			
No. Auctions	22.6	12	24.5
No. Firms	25.9	15	31.0
Asphalt			
No. Auctions	16.2	11	18.0
No. Firms	11.6	7	15.0

Note: Each panel restricts to counties with at least ten auctions in that category. Statistics are computed across counties within panel. Large auctions are defined as those with a winning bid greater than $1 million (roughly the median winning bid). "No. Auctions" counts distinct auctions (conditional on at least 10; this drops roughly 40% of counties. "No. Firms" counts distinct bidders participating in those auctions.

Table A.13: County-level number of bidders

Bid Rotation Simple bid rotation is a straightforward method of collusion. Chassang et al. (2022) propose a test for bid rotation based on a regression discontinuity design. The outcome is a variable plausibly related to rotation, such as firm backlog. The running variable is $\Delta_{i,t} = b_{i,t} - \wedge\mathbf{b}_{-i,t}$, the difference between each bid $b_{i,t}$ and the lowest other bid in the same auction.

Table A.14 present results from this test, using backlog as the outcome. Backlog is defined as the total value of projects won in the prior 90 or 180 days, following Chassang et al. (2022). For the vast majority of states, the null of no treatment effect at the threshold cannot be rejected. Only three states—Mississippi, Arizona, and Wisconsin—reject the null at the 5% level for 90-day backlog, and no state does so for 180-day backlog. In both Mississippi and Arizona, the estimated discontinuity is positive, indicating that firms with *more* recent backlog are more likely to win. This pattern is inconsistent with standard bid-rotation schemes, which predict negative persistence in wins. Therefore, these few rejections are most plausibly explained by multiple hypothesis testing rather

[58] Further, Project type is not a natural market definition—firms typically bid across types.

than collusion.

Outcome	10th pct.	Median	90th pct.
Backlog (90 days)	0.069	0.382	0.909
Backlog (180 days)	0.091	0.374	0.857

Note: The table reports quantiles of state-level p-values from regression discontinuity tests following Kawai, Nakabayashi, Ortner, and Chassang (2022). Estimates use local linear regression with robust bias-corrected inference. The quantiles summarize the distribution of p-values across states.

Table A.14: Quantiles of RD p-values across states

Independence Tests Bajari and Ye (2003) propose a conditional independence test to distinguish competitive from collusive bidding. Under competition, bids submitted by two firms should be uncorrelated across auctions after conditioning on auction and firm characteristics. The test computes pairwise correlations in bids and uses a Fisher Z-transformation to test independence for each firm pair.

I implement this test using homogenized bids for 2023, restricting attention to firm pairs that bid together in at least four auctions, as in Bajari and Ye (2003). Because the homogenization does not control for firm-specific characteristics, rejection of independence may reflect collusion, persistently correlated firm heterogeneity or failure in the homogenization procedure. I correct for multiple hypothesis testing using the Benjamini–Hochberg procedure. After correction, approximately 5.2 percent of firm pairs reject the null of independence at the 5 percent level. Since the share of rejections is small relative to the number of tests conducted, these results also do not point to widespread collusive behavior.

C.2 Robustness of homogenization regression specification

Model	Median RMSE	IQR RMSE	Pooled RMSE
Baseline	0.250	0.092	0.271
+ unit covariates	0.223	0.065	0.220
+ item covariates	0.232	0.078	0.244

Table A.15: Cross-validated root mean squared error (RMSE) for three regression specifications. Median and inter-quartile range (IQR) are calculated across states; the pooled RMSE weights observations equally. The baseline specification includes fixed effects for engineering type and AI-derived project class, county-level terrain ruggedness and rurality, and log total inputs measured in tons. The "+ unit covariates" specification (preferred, corresponding to regression 8) adds log total material quantities aggregated by measurement unit—tons, cubic yards, linear feet, square yards, and counts of "each"—and log the number of distinct bid-schedule items; all physical quantities are converted to common units (e.g., pounds to tons). The "+ item covariates" specification further includes individual item quantities for every item used in at least 25% of projects, along with log the number of distinct items and the count of "rare" items below this threshold. Item effects are regularized with ridge penalties of $\lambda = 0.5$ on the item coefficients. All specifications ridge penalities of $\lambda = 1$ on the auction-level intercept, and 0 on the baseline covariates.

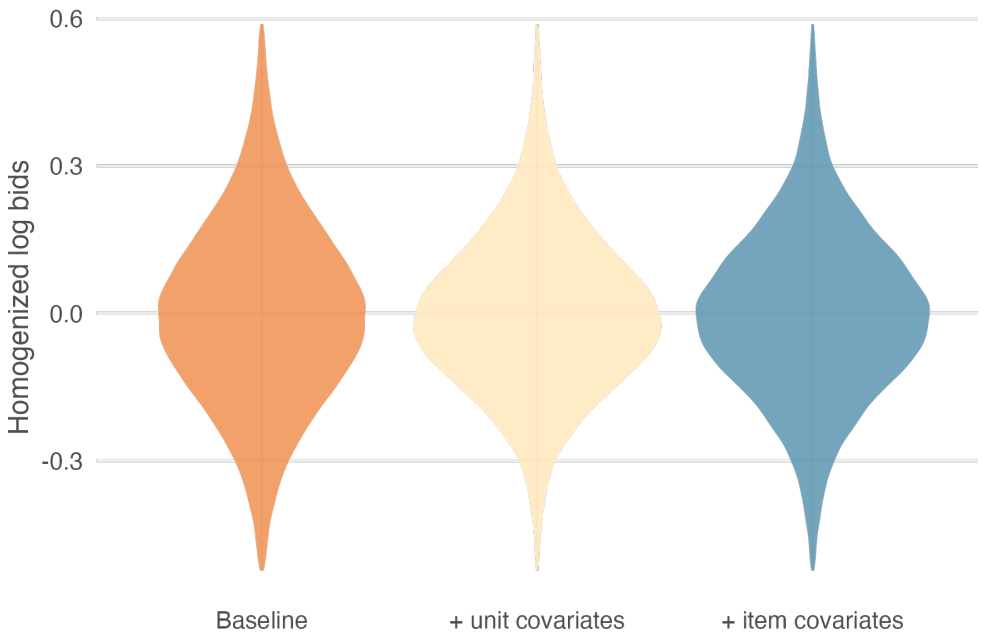


Figure A.13: Distribution of homogenized log bids by specification

Note: Figure plots violin densities of state-demeaned log bids for the three regression specifications defined in Table A.15. Observations beyond the 0.5th and 99.5th percentiles of the residual distribution are dropped before plotting.

C.3 Robustness of number of bidder prediction

	Dependent variable:					
	N		log bid	log homogenized bid		
	(1)	(2)	(3)	(4)	(5)	(6)
E[N], multinomial	1.639***				−0.032***	
	(0.015)				(0.002)	
E[N], xgboost		1.239***				−0.020***
		(0.004)				(0.001)
N			−0.0005	−0.010***		
			(0.003)	(0.001)		
state FE	yes	yes	yes	yes	yes	yes
Projected R-squared	0.201	0.719	0	0.009	0.004	0.01
Observations	47,431	47,431	46,952	46,952	44,793	44,793
Note:						*p<0.1; **p<0.05; ***p<0.01

Table A.16: Columns (1) and (2) show the correlation between the predicted number of bidders from two models and the actual number of bidders. Columns (3) through (6) show the correlations between the number of bidders—either actual or predicted—and both actual and homogenized bids. All models include the same covariates: fixed effects for engineering type and AI-derived project class, county-level terrain ruggedness and rurality, total input quantities measured in tons, individual quantities for all bid items present in at least 25% of projects, the log number of distinct items, and the count of "rare" items below this frequency threshold. The multinomial model is a multinomial logistic regression with a ridge penalty. The XGBoost model is a gradient-boosted decision-tree model.

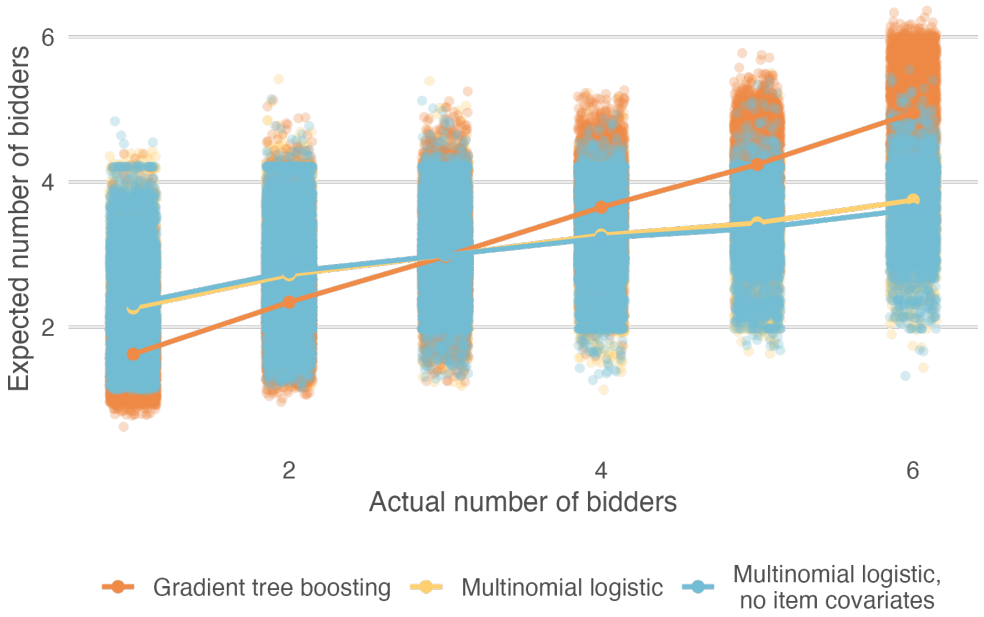


Figure A.14: Number of bidder prediction by different models

Note: This figure compares actual and predicted bidder counts using the models shown in Table A.16. It also includes predictions from a third model—identical to the multinomial logistic model but excluding bid schedule item covariates—to illustrate that omitting these covariates has only a minor impact on predictions. In contrast, switching from the penalized multinomial logistic model to the XGBoost model substantially alters predicted bidder counts.

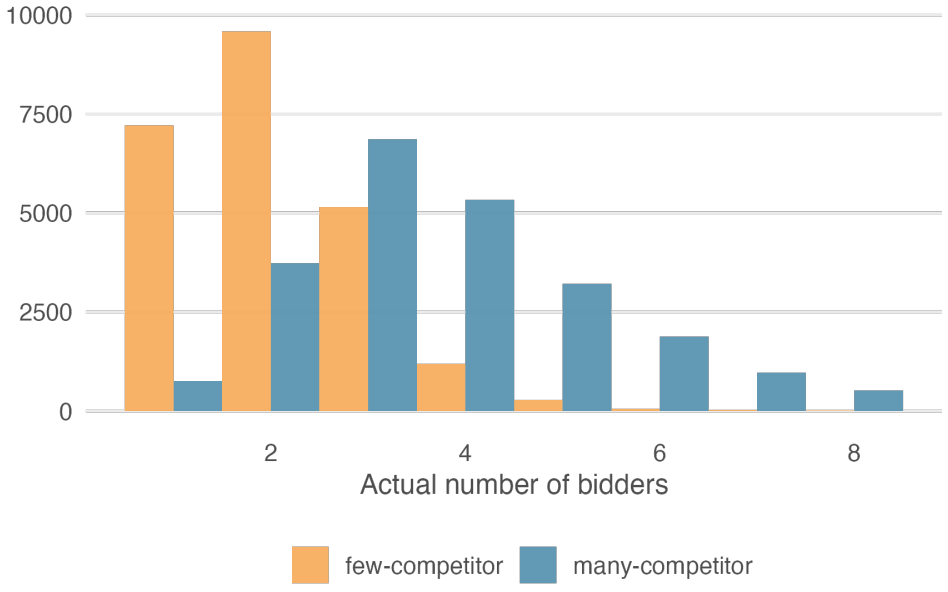


Figure A.15: Auction clusters based on predicted number of bidders

Note: Figure shows the distribution of auctions across the actual number of bidders – "few-competitor" and "many-competitor" – by the predicted bidder count clusters. Clusters are based on a median split of the predicted number of bidders from the gradient tree boosting (XGBoost) algorithm. Details on the algorithm are given in Table A.16.

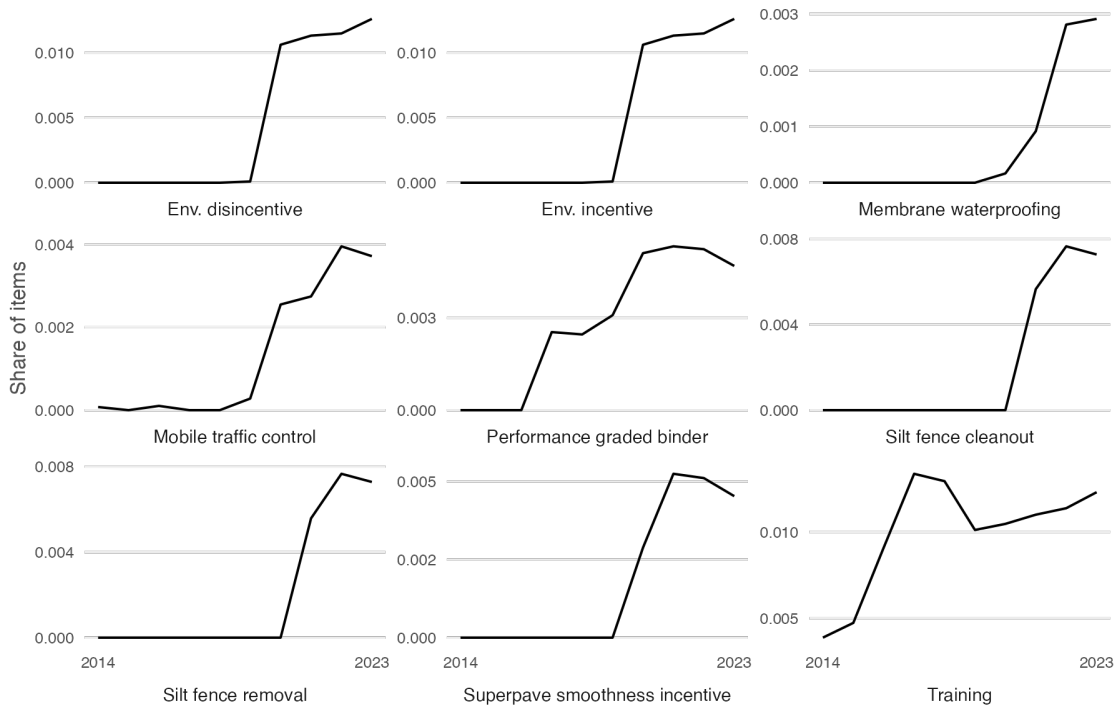


Figure A.16: Most Rapidly Growing Input Items, Nebraska 2014–2023

Note: Figure shows trends in Nebraska pay items, 2014–2023. Each panel plots the yearly share of items for the items with the highest share growth over the period. Shares are defined as the item's count divided by the total items that year. Mobilization is omitted as a category, as that represents a lump sum amount and is not an input used to control for project characteristics. Over the full period, the average unique items per project rose rose by 19% from 54 to 64.

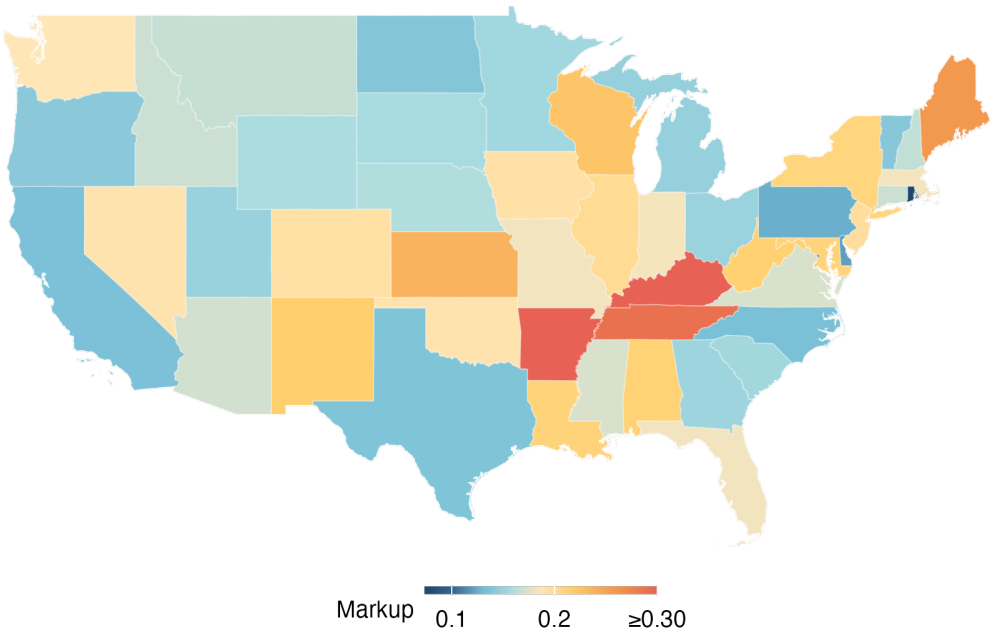


Figure A.17: Median Markup Across States

Note: Figure shows the median winning-bid markup by state from 2019 to 2023. Each state's color represents the median markup among winning bids on all roadwork projects over the period. Projects with negative cost estimates are dropped from the sample. Projects with a markup above .3 are grouped together in the highest-color category. The highest median markup estimate is for Kentucky, with an estimate of .54.

C.4 MCMC implementation and diagnostics

I estimate the parameters of the entry model in Section 7.2 using an independence Metropolis–Hastings algorithm. The likelihood is given in Equation 9. I run four independently seeded chains, initialized at dispersed points along the two directions in which a local approximation to the posterior varies most. Each chain produces 9,500 retained posterior draws. The first 500 iterations of the initial block are discarded, so the computation involves 10,000 transitions per chain. All 9,500 retained draws per chain, or 38,000 draws in total, are used for posterior inference.

I place independent priors on the model parameters. Each cluster-specific entry-cost intercept κ_c has a log-normal prior chosen to have a central 95 percent interval of $[50, 1500]$. Each cluster-specific common-shock standard deviation ρ_c has a half-normal prior with scale 250, and the firm-specific shock standard deviation σ has a half-normal prior with scale 500. The coefficients on standardized five-year bidding experience and standardized state breadth have independent $N(0, 100^2)$ priors.

The coefficient on standardized log project scale is restricted to be nonnegative and has a half-normal prior with scale 250. All cost parameters are measured in thousands of 2023 dollars. These priors impose the required parameter support and regularize economically implausible regions while retaining substantial prior dispersion.

The independence proposal is constructed as follows. Before running the chains, I locate the posterior mode by numerical optimization from three starting values. I calculate the Hessian at the best solution and use its inverse to approximate the posterior variances and correlations near the mode. Based on this approximation, I construct a fixed mixture of four eight-dimensional Student-t proposal distributions. Each component has seven degrees of freedom and is centered at the posterior mode. The components share the same correlation structure but have covariance multipliers of 0.5, 1, 2, and 4. At each iteration, one component is selected with probabilities 0.15, 0.45, 0.30, and 0.10, respectively, and a complete eight-parameter candidate is drawn independently of the chain's current state. The candidate is accepted or rejected using the independence Metropolis–Hastings acceptance probability, which accounts for the proposal density at both the candidate and current states. Thus, this is an independence proposal rather than a random-walk proposal. The proposal remains fixed throughout warmup and sampling.

I assess MCMC convergence and simulation precision using the standard measures of $\hat{R}$, effective sample size (ESS), and Monte Carlo standard error (MCSE). For $\hat{R}$, I report the maximum of the rank-normalized and folded split statistics. In its basic form, $\hat{R} = \sqrt{\frac{\frac{N-1}{N}W+\frac{1}{N}B}{W}}$, where W is the average within-chain variance, B is the scaled variance of the chain means, and N is draws per chain. Values close to one indicate that the chains are sampling the same stationary distribution. I use the split, rank-normalized and folded versions, as recommended in modern MCMC diagnostics. Values close to one indicate that the chains are sampling the same stationary distribution.

I assess simulation precision using rank-normalized, split-chain effective sample sizes (ESS). Bulk ESS measures the effective number of independent draws available for estimating the central portion of the posterior distribution. Tail ESS is calculated separately for the 5th and 95th posterior quantiles, and I report the smaller value, measuring precision in the more weakly sampled tail. I also report the Monte Carlo standard error relative to the posterior standard deviation, which is interpreted as simulation error relative to posterior uncertainty.

Table A.17 reports these diagnostics, and Figure A.18 shows the four chains. The maximum $\hat{R}$ is 1.009, and the minimum bulk ESS is 549; thus, all parameters satisfy the commonly used guidelines of $\hat{R} < 1.01$ and bulk ESS above 400. Tail ESS exceeds 400 for all parameters except ρ_{few}, for which it is 137, and α_{exp}, for which it is 382, indicating less precise estimation of their posterior tails. The largest ratio of the Monte Carlo standard error to the posterior standard deviation is 0.053. Thus, the chains exhibit good convergence and computational precision overall, with more limited precision in the lower tail of ρ_{few}.

Parameter	R-hat	Bulk ESS	Tail ESS	MCSE mean / SD
κ_{few}	1.003	1335	687	0.027
κ_{many}	1.004	1721	3264	0.024
ρ_{few}	1.009	549	137	0.053
ρ_{many}	1.004	1985	945	0.022
α_{exp}	1.004	1130	382	0.030
α_{scope}	1.001	3176	2999	0.018
β	1.003	2080	1486	0.024
σ	1.002	965	592	0.032

Note: Table reports MCMC convergence diagnostics for each estimated parameter. R-hat is the maximum split $\widehat{R}$ statistic across chains. Bulk ESS and Tail ESS report effective sample sizes for the center and tails of the posterior distribution, respectively. MCSE / SD is the Monte Carlo standard error of the posterior mean divided by the posterior standard deviation. Values of $\widehat{R}$ close to one, large effective sample sizes, and small MCSE / SD indicate good convergence and simulation precision.

Table A.17: MCMC convergence diagnostics

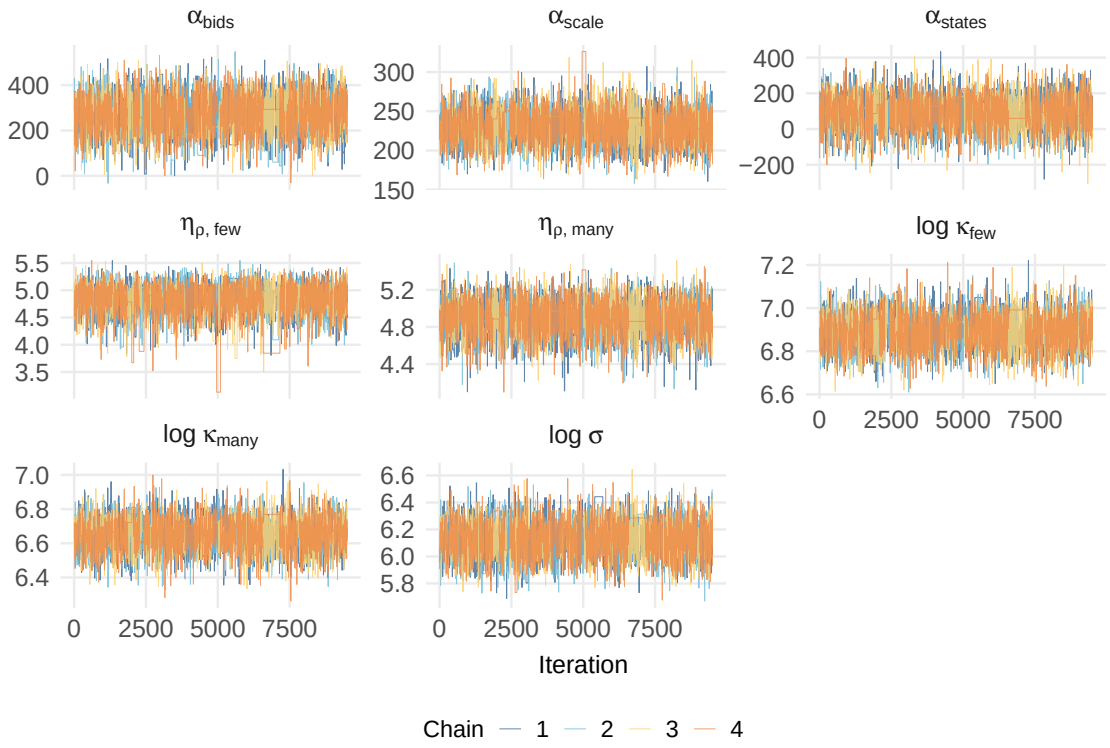


Figure A.18: MCMC Trace Plots

Note: Figure shows trace plots for the four MCMC chains for each estimated parameter. Each panel plots the retained post-warmup draws over iteration, with colors denoting separate chains. The plots show κ and σ in logs and ρ in its transformed parameterization, matching the coordinates used for sampling; the two elements of α are shown with the signs used in the text. For visual clarity only, the trace plots display every fifth retained draw.

Statistic	P75: smaller pool	P90: main model	P100: larger pool
Entrant pool			
Mean potential bidders	13.2	18.1	31.7
Median potential bidders	11.0	16.0	31.5
Auctions requiring support additions	3 (1.1%)	2 (0.7%)	0
Mean entry cost, first three entrants ($000)	120.9	111.5	42.0
MCMC diagnostics			
Maximum $\widehat{R}$	1.0041	1.0071	1.0145
Minimum bulk ESS	1298.7	548.6	317.6
Minimum tail ESS	575.1	136.7	98.0
Maximum MCSE/SD	0.0345	0.0533	0.0659

Note: This table reports sensitivity of the estimated entry costs to the definition of the potential-entrant pool. The baseline specification, P90, defines each firm's distance cutoff using the 90th percentile of that firm's historical bidding distances within Illinois. The P75 specification instead uses the firm's 75th percentile, while the P100 specification uses the firm's maximum observed bidding distance within Illinois, shifting pool size as shown in the first two rows. "Auctions requiring support additions" reports the number of auctions for which one additional nonbidding firm must be added so that the potential-entrant pool contains at least $N + 1$ firms. While the P75 and P90 specifications estimate similar entry costs, the P100 specification, with a substantially larger potential-entrant pool, produces different estimates. However, the P100 specification does not satisfy the reported convergence and precision criteria, indicating that the model has difficulty fitting this broader entrant-pool specification.

Table A.18: MCMC sensitivity to the potential-entrant pool

D Technical Appendix

D.1 Optimal bidding equation

The first order equation for the bidding firm is i

$$\begin{aligned}\frac{\partial \mathbb{E}\Pi}{\partial \beta} = (\beta - c_i)\Big[\sum_n \Pr(N = n|X)\,(n-1)\big(1 - F(\beta^{-1})\big)^{n-2}\Big(- f(\beta^{-1})\,\tfrac{1}{\beta'}\Big)\big(1 - H(\beta|X)\big) \\ + \sum_n \Pr(N = n|X)\big(1 - F(\beta^{-1})\big)^{n-1}\big(- h(\beta|X)\big)\Big] + \\ + \sum_n \Pr(N = n)\big(1 - F(\beta^{-1})\big)^{n-1}\big(1 - H(\beta|X)\big) = 0.\end{aligned}$$

Solving for the bid strategy gives

$$\beta(c_i) = c_i + \frac{\sum_n \Pr(N = n|X)\,\big(1 - F(\beta^{-1}|X)\big)^{n-1}\,(1 - H(\beta|X))}{\sum_n \Pr(N = n|X)\,\big(1 - F(\beta^{-1}|X)\big)^{n-2}\left((n-1)\frac{f(\beta^{-1}|X)}{\beta'}(1 - H(\beta|X)) + \big(1 - F(\beta^{-1}|X)\big)\,h(\beta|X)\right)}$$

D.2 Proof of equation 7

The proof closely follows the proof of Krasnokutskaya, 2011 Proposition 1. I show that the proposed bidding function satisfies the conditions for equilibrium. Note that I have already assumed equilibrium uniqueness.

Suppose $z_i = c_i \cdot y$ where $y \in [\underline{y}, \bar{y}]$ is a common multiplicative shock observed by all bidders in the auction. Let $\alpha(c_i)$ denote the symmetric equilibrium bidding strategy in the benchmark case $y = 1$, with distribution F of c_i, density f, entry probabilities $p_N(n)$ and government reserve $r \sim H(a)$.

The first order condition for the $y = 1$ auction over is

$$a_i = c_i + \frac{\sum_n P(n)\left(1 - F(\alpha^{-1}(a_i))\right)^{n-1}\left(1 - H(a_i)\right)}{\sum_n P(n)\left[(n-1)\left(1 - F(\alpha^{-1}(a_i))\right)^{n-2} f(\alpha^{-1}(a_i))\,\frac{1}{\alpha'\left(\alpha^{-1}(a_i)\right)}\left(1 - H(a_i)\right) + \left(1 - F(\alpha^{-1}(a_i))\right)^{n-1} h(a_i)\right]}$$

Now consider a general auction with cost $z_i = c_i \cdot y$. The first order condition is

$$b_i = c_i y + \frac{\sum_n P(n)\left(1 - G\left(\beta^{-1}(b_i)\right)\right)^{n-1}\left(1 - K(b_i)\right)}{\sum_n P(n)\left[(n-1)\left(1 - G\left(\beta^{-1}(b_i)\right)\right)^{n-2} \cdot g\left(\beta^{-1}(b_i) \cdot \frac{1}{\beta'\left(\beta^{-1}(b_i)\right)}\right)\left(1 - K(b_i)\right) + \left(1 - G\left(\beta^{-1}(b_i)\right)\right)^{n-1} k(b_i)\right]}.$$

I hypothesize that the bidder strategy is $\beta(z_i) = y \cdot \alpha\left(\frac{z_i}{y}\right)$. I now show that this satisfies the differential equation. First note that $\frac{\partial \beta(z_i)}{\partial z_i} = y \cdot \alpha'\left(\frac{z_i}{y}\right) \cdot \frac{1}{y} = \alpha'\left(\frac{z_i}{y}\right)$. Next, $\beta^{-1}(b_i) = y \cdot \alpha^{-1}\left(\frac{b_i}{y}\right)$. Finally, the distributions can be written:

$$G\left(\beta^{-1}(b_i)\right) = F\left(\alpha^{-1}\left(\frac{b_i}{y}\right)\right),$$
$$g\left(\beta^{-1}(b_i)\right) = \frac{1}{y} f\left(\alpha^{-1}\left(\frac{b_i}{y}\right)\right).$$

Plugging these definitions into the general auction FOC gives:

$$b_i = c_i\, y + y \cdot \frac{\sum_n P(n)\left(1 - F\left(\alpha^{-1}\left(\frac{b_i}{y}\right)\right)\right)^{n-1}\left(1 - K\left(y \cdot \alpha\left(\frac{z_i}{y}\right)\right)\right)}{\sum_n P(n)\left[(n-1)\left(1 - F\left(\alpha^{-1}\left(\frac{b_i}{y}\right)\right)\right)^{n-2} f\left(\alpha^{-1}\left(\frac{b_i}{y}\right)\right)\frac{1}{\alpha'\left(\alpha^{-1}\left(\frac{b_i}{y}\right)\right)}\left(1 - K\left(y \cdot \alpha\left(\frac{z_i}{y}\right)\right)\right) + \left(1 - F\left(\alpha^{-1}\left(\frac{b_i}{y}\right)\right)\right)^{n-1} k\left(y \cdot \alpha\left(\frac{z_i}{y}\right)\right)\right]}$$

The final requirement for equality is on the secret reserve price, specifically that $K(y \cdot a_i) = H(a_i)$ (and consequently, $k(b_i) = \frac{1}{y} h(a_i)$). In words, when the cost and bids are shifted by y, the reserve must also shift by y, preserving scale invariance. One solution is to simply assume $r = y \cdot \rho$ where ρ corresponds to the $y = 1$ auction. In practice, I make the stronger assumption that the reserve is drawn from a distribution that is a fixed multiple of the bid distribution, so as bids scale with y the reserve distribution scales proportionally by the same constant factor.

D.3 Counterfactual estimation

Step 1: Rearrange first order equation Let $A(c) = \sum_n P(n)\,(1-F(c))^{n-1}$ and $G(c) = \sum_n P(n)(n-1)\,f(c)\,(1-F(c))^{n-2}$. Then equation 5 can be written

$$
\begin{aligned}
b - c &= \frac{A(c)\,(1-H(b))\,\beta'}{G(c)\,(1-H(b)) + A(c)\,h(b)\,\beta'}, \\
\implies A(c)h(b)(b-c)\,\beta' + (1-H(b))(b-c)G(c) &= A(c)(1-H(b))\,\beta', \\
\implies \beta'\big[A(c)h(b)(b-c) - A(c)(1-H(b))\big] &= -(1-H(b))(b-c)G(c), \\
\implies \beta' &= \frac{-(1-H(b))(b-c)G(c)}{A(c)\left[h(b)(b-c) - (1-H(b))\right]}, \\
&= (b-c)\,\frac{G(c)}{A(c)}\,\frac{1-H(b)}{1-H(b)-h(b)(b-c)},
\end{aligned}
$$

where $b = \beta(c)$. This is an explicit first order ordinary differential equation.

Step 2: Write the initial condition I assume the secret reserve price has a mass point at the right upper bound of the support $\bar{c}$, rather than a right tail past this point. Empirically, this makes no difference beyond the starting point of the ODE. At $\bar{c}$, the bid is $\bar{b} = \beta(\bar{c}) = \bar{c}$. I use this point as the initial value and iterate left along the empirical support of c to trace out the full bid function.

In a handful of (state, year, cluster) samples, the solver fails because the denominator in the ODE approaches zero at the boundary. To address this, I shift the starting point to $(\bar{c} - \epsilon, \bar{b})$, where ϵ is less than .5% of costs and $\bar{b}$ is a first order Taylor approximation of $\beta(\bar{c} - \epsilon)$, as the derivative at the boundary is zero.

Step 3: Numerically solve I estimate a separate solution curve for each (state, year, cluster) using $\beta' = (b-c)\,\frac{G(c)}{A(c)}\,\frac{1-H(b)}{1-H(b)-h(b)(b-c)}$ and a Rosenbrock solution method for stiff differential equations, provided by the Matlab function 'ode23s'.